STAT201 · STATISTICS · FOR THE INSTRUCTOR

Answers and detailed solutions to the problems of lectures 1--15

All numbers were recomputed by software; solutions use the same formulas and rounding rules as the lectures

How to use the key

The problems in each lecture are marked by level: **[A]** basic, **[B]** standard, **[C]** advanced. Below are the statements and full solutions.

- Decimals are written with a point (as on a calculator, in Excel and in R).
- Quartiles are computed by the median-of-halves method (in Excel `QUARTILE.INC` gives 26.5 and 37 for the data of lecture 1; `QUARTILE.EXC` gives 25.5 and 39). Agree on the method with students.
- Answers read from distribution tables may differ from software values in the third digit: accept an answer within ± 0.002 for probabilities and ± 0.01 for critical values.
- The sample variance is computed with divisor $n-1$.
- Critical values and p-values are given from the exact distributions (Python/`scipy`); the appendix tables give the same numbers rounded.

ANSWERS AND SOLUTIONS · LECTURE 1

Data, Descriptive Statistics and Visualization

Problem 1 [A] For each variable state the measurement scale and which measures of centre (mode, median, mean) are admissible:

(a) employee number;
(b) years of education;
(c) satisfaction with a service on a scale of 1–5;
(d) monthly salary in UZS;
(e) region of residence;
(f) year of birth;
(g) air temperature in °C;
(h) place achieved in a competition (1st, 2nd, …).

SOLUTION

Variable	Scale	Admissible measures of centre
(a) employee number	nominal	only the mode (and it is hardly meaningful)
(b) years of education	ratio (0 years = no education)	mode, median, mean
(c) satisfaction 1–5	ordinal	mode, median; the mean is an assumption
(d) salary in UZS	ratio	all three
(e) region	nominal	mode
(f) year of birth	interval (arbitrary zero)	all three; “twice as old” makes no sense
(g) temperature in °C	interval	all three
(h) place in a competition	ordinal	mode, median

Problem 2 [A] True or false? Justify briefly.

(a) The median does not depend on the size of the extreme values.
(b) If the mean is greater than the median, the distribution is, as a rule, right-skewed.
(c) The sum of deviations from the median is always zero.
(d) The variance can be negative.
(e) If all values are multiplied by 2, the standard deviation doubles and the coefficient of variation does not change.
(f) The line inside the box of a box plot shows the mean.

SOLUTION

(a) **True.** The median is determined by the order of the values, not their size: if the largest value is increased a thousandfold, the median does not change.
(b) **True** (as a rule): a long right tail pulls the mean to the right more than the median.
(c) **False.** It is the sum of deviations from the *mean* that equals zero, the “balance point” property (Lecture 1, section 4.1).

(d) **False.** The variance is a sum of squares divided by a positive number, so it is non-negative; it equals zero only if all values are the same.
(e) **True.** By the linear transformation rule $s_y = 2s_x$ and $\bar{y} = 2\bar{x}$, so CV does not change.
(f) **False.** The line shows the median.

Problem 3 [B] A courier's number of deliveries over seven days: 12, 15, 15, 18, 22, 25, 61. Find the mean, median, mode, range, variance, standard deviation and coefficient of variation. Use the $1.5 \times$ IQR rule to check for outliers. Which measure of centre would you publish and why? Answer in one sentence.

SOLUTION $n = 7$, $\sum x_i = 168$.
Mean $\bar{x} = 168/7 = 24$. The median is the value in position $(7+1)/2 = 4$: Me $= 18$. Mode $= 15$. Range $= 61 - 12 = 49$.
Deviations: $-12, -9, -9, -6, -2, +1, +37$; squares: $144, 81, 81, 36, 4, 1, 1369$; $\sum (x_i - \bar{x})^2 = 1716$. $s^2 = 1716/6 = 286$, $s = \sqrt{286} = 16.91$, $CV = 16.91/24 = 70.5\%$.
Quartiles: the lower half $(12, 15, 15)$ gives $Q_1 = 15$; the upper $(22, 25, 61)$ gives $Q_3 = 25$; IQR $= 10$. Fences: $15 - 15 = 0$ and $25 + 15 = 40$. Since $61 > 40$, the value 61 is an outlier.
Conclusion: publish the median: "a typical day is 18 deliveries", because the mean of 24 is inflated by one unusually busy day (without it the mean is $107/6 = 17.83$, almost the same as the median).

Problem 4 [B]

(a) A bank issued three loans: 50 million UZS at 18%, 30 million at 22% and 20 million at 24%. Find the average interest rate of the portfolio (simple and weighted) and explain the difference.
(b) A student got 85 points in a 5-credit course, 70 points in a 4-credit course and 90 points in a 3-credit course. Find the weighted average score.

SOLUTION

(a) Simple: $(18 + 22 + 24)/3 = 21.33\%$. Weighted (weight = loan amount): $\dfrac{50 \cdot 18 + 30 \cdot 22 + 20 \cdot 24}{100} = \dfrac{900 + 660 + 480}{100} = 20.4\%$. The weighted rate is lower because the largest loan was issued at the lowest rate: most of the money works at 18%. The weighted rate is what shows the bank's real income from the portfolio.
(b) $\dfrac{85 \cdot 5 + 70 \cdot 4 + 90 \cdot 3}{5 + 4 + 3} = \dfrac{425 + 280 + 270}{12} = \dfrac{975}{12} = 81.25.$

Problem 5 [B] Monthly rent of 15 apartments (million UZS): 2.4, 2.6, 2.8, 3.0, 3.1, 3.2, 3.5, 3.6, 3.8, 4.0, 4.2, 4.5, 4.8, 5.0, 9.5.

(a) Find the five-number summary and the IQR.
(b) Find the outlier fences and identify the outliers.
(c) Describe the box plot: where do the whiskers end?
(d) Compare the mean and the median. What do they say about the shape of the distribution? Which measure would you report?

SOLUTION $n = 15$, $\sum x_i = 60$.
(a) Min $= 2.4$; the median is the 8th value, Me $= 3.6$. The lower half (the first 7 values) has

as its median the 4th value: $Q_1 = 3.0$; for the upper half (the last 7) $Q_3 = 4.5$; max $= 9.5$. IQR $= 4.5 - 3.0 = 1.5$.

(b) $1.5 \cdot 1.5 = 2.25$. Fences: $3.0 - 2.25 = 0.75$ and $4.5 + 2.25 = 6.75$. The value $9.5 > 6.75$ is an outlier; the others are inside the fences.

(c) The box runs from 3.0 to 4.5 with a line at 3.6; the left whisker to 2.4, the right to 5.0 (the largest value inside the fence); a separate point at 9.5. The median is closer to the lower edge of the box and the right whisker is longer — right skew.

(d) Mean $= 60/15 = 4.0$ is greater than the median 3.6 — right skew: the mean is pulled to the right by the value 9.5 (without it the mean is 3.61). We report the median of 3.6 million and the IQR of 1.5 million, and separately point out the apartment at 9.5 million.

Problem 6 [B] Number of orders at a coffee shop over 30 days: 14, 18, 19, 21, 22, 23, 24, 24, 25, 26, 27, 27, 28, 29, 29, 30, 31, 31, 32, 33, 34, 35, 36, 37, 38, 40, 42, 45, 49, 58.

(a) Build a frequency table with intervals $[10, 20), [20, 30), \ldots, [50, 60)$: frequency, relative frequency, cumulative frequency.

(b) On what share of days were there fewer than 30 orders?

(c) State the modal interval.

(d) Find the mean and median from the raw data and compare them. What does this say about the shape?

SOLUTION

(a)

Interval	f	f/n	cumul.
[10,20)	3	10.0%	3
[20,30)	12	40.0%	15
[30,40)	10	33.3%	25
[40,50)	4	13.3%	29
[50,60)	1	3.3%	30

Check: $3 + 12 + 10 + 4 + 1 = 30$.

(b) The cumulative frequency for the intervals up to 30: $15/30 = 50\%$.

(c) The modal interval is $[20, 30)$, $f = 12$.

(d) $\sum x_i = 927$, $\bar{x} = 927/30 = 30.9$. The median is the average of the 15th and 16th values: $(29 + 30)/2 = 29.5$. The mean is slightly greater than the median — weak right skew (a tail of the values 45, 49, 58).

Problem 7 [B]

(a) The average salary in firm A is 2.4 million UZS with a standard deviation of 0.6 million; in firm B it is 5.0 million with a standard deviation of 1.0 million. Where are salaries more homogeneous?

(b) The return on stock X: mean 8%, standard deviation 12%; on stock Y: mean 5%, standard deviation 6%. Which stock has less risk per unit of return?

SOLUTION

(a) $CV_A = 0.6/2.4 = 25\%$, $CV_B = 1.0/5.0 = 20\%$. Although the absolute deviation is larger in B, relative to their own scale the salaries in firm B are more homogeneous.

(b) $CV_X = 12/8 = 150\%$, $CV_Y = 6/5 = 120\%$. Stock Y has less risk per unit of return. (The coefficient of variation is sensible only while the mean is positive and far from zero.)

Problem 8 [C]

(a) Prove that $\sum(x_i - \bar{x}) = 0$.

(b) The salaries of five employees (million UZS): 2.0, 2.5, 3.0, 3.5, 4.0. Find $\bar{x}$, s and CV. Then each salary is raised by 10% and a bonus of 0.2 million is added: $y = 1.1x + 0.2$. Find $\bar{y}$, s_y, CV_y and explain why CV changed.

SOLUTION

(a) $\sum(x_i - \bar{x}) = \sum x_i - \sum \bar{x} = n\bar{x} - n\bar{x} = 0$, since $\bar{x}$ is a constant and $\sum x_i = n\bar{x}$ by the definition of the mean.

(b) $\bar{x} = 15/5 = 3.0$. Squared deviations: $1,\ 0.25,\ 0,\ 0.25,\ 1$; sum 2.5; $s^2 = 2.5/4 = 0.625$; $s = 0.79$; $CV = 0.79/3.0 = 26.35\%$.
$\bar{y} = 1.1 \cdot 3.0 + 0.2 = 3.5$; $s_y = 1.1 \cdot 0.79 = 0.87$; $CV_y = 0.87/3.5 = 24.85\%$.
Explanation: the factor 1.1 stretches the centre and the spread equally, while the bonus 0.2 shifts the centre without changing the spread. The denominator of CV grew, the numerator received no shift, so the relative spread decreased.

Problem 9 [C]

Monthly incomes of 10 households (million UZS): 2.1, 2.4, 2.6, 2.8, 3.0, 3.1, 3.4, 3.9, 4.5, 18.2. The ministry reports "average income 4.6 million", the trade union "typical income 3.05 million". Both use the same data honestly.

(a) Verify both numbers.

(b) How many of the ten households have an income below the "average"?

(c) Check for outliers.

(d) Which number should a family planning its budget use, and why?

SOLUTION

(a) $\sum x_i = 46.0$, $\bar{x} = 4.6$. The median is the average of the 5th and 6th values: $(3.0 + 3.1)/2 = 3.05$. Both numbers are correct.

(b) Nine out of ten (all except 18.2) have an income below 4.6. The "average income" does not describe a typical family.

(c) $Q_1 = 2.6$, $Q_3 = 3.9$, IQR $= 1.3$, $1.5 \cdot 1.3 = 1.95$; the upper fence is $3.9 + 1.95 = 5.85$. The value $18.2 > 5.85$ is an outlier.

(d) The median: it describes the family in the middle of the distribution and is not distorted by one very high income. Without the outlier the mean is 3.09 and the median is 3.0: they almost coincide, which confirms that the discrepancy is caused by the outlier.

Problem 10 [C]

A company reports that the share of satisfied customers rose from 94% to 97%. In the bar chart the y-axis starts at 93%.

(a) By how many percentage points and by what relative percentage did the share grow?

(b) How many times taller does this year's bar look than last year's?

(c) How would you fix the graph? Write a caption for the corrected graph.

SOLUTION

(a) Growth by $97 - 94 = 3$ percentage points, or $3/94 = 3.2\%$ in relative terms.

(b) The bar heights from the start of the axis: $97 - 93 = 4$ and $94 - 93 = 1$, i.e. the bar looks **4 times taller** (by 300%) while real growth is 3.2%.

(c) Start the axis at zero (the bars will be almost equal, which honestly reflects the small growth) or show only the numbers with an explicit note "axis does not start at zero". Caption: "The share of satisfied customers rose by 3 p.p., to 97%".

Problem 11 [C] Hours of preparation x and score y for eight students: $(1, 45), (2, 52), (3, 58), (4, 61), (5, 70), (6, 74), (7, 79), (8, 60)$.

(a) Draw the scatter plot.

(b) Describe it in four words: direction, form, strength, unusual points.

(c) What will you do with the point $(8, 60)$ before drawing conclusions?

(d) Can we claim that more hours of preparation *cause* a higher score?

SOLUTION

(b) **Direction** — positive: for the first seven students the score grows with the number of hours. **Form** — roughly linear (an increase of 5–9 points per hour). **Strength** — strong for the seven points (for them $r = 0.996$; for all eight $r = 0.76$: one point noticeably weakens the relationship). **Unusual point** — $(8, 60)$: eight hours of preparation, but a lower score than the student with three hours.

(c) Check: a typo in the data? Illness or fatigue on exam day? A different exam format? The point must not be removed silently; report the results with and without it.

(d) No. A relationship does not prove causation: perhaps more motivated students both study more and do better in the exam. We shall return to this question in week 13.

ANSWERS AND SOLUTIONS · LECTURE 2

Foundations of Probability Theory

Problem 1 [A] A questionnaire has two questions, each of which can be answered "yes" (Y) or "no" (N).

(a) Write down the sample space S.
(b) Write down the event "at least one answer is yes". How many outcomes does it contain?
(c) Find its probability, assuming all outcomes are equally likely.

SOLUTION

(a) $S = \{\text{YY},\ \text{YN},\ \text{NY},\ \text{NN}\}$, 4 outcomes.
(b) $\{\text{YY}, \text{YN}, \text{NY}\}$, 3 outcomes.
(c) $P = 3/4 = 0.75$. (Check via the complement: $1 - P(\text{NN}) = 1 - 1/4 = 0.75$.)

Problem 2 [A]

(a) $P(\text{late}) = 0.30$. Find $P(\text{not late})$.
(b) $P(A) = 0.40$, $P(B) = 0.30$, $P(A \cap B) = 0.15$. Find $P(A \cup B)$.
(c) It is known that $P(A) = 0.20$ and $P(A \cap B) = 0.35$. What can you say about these numbers?
(d) Two dice are rolled. Find the probability that at least one six came up (in two ways).

SOLUTION

(a) $1 - 0.30 = 0.70$.
(b) $0.40 + 0.30 - 0.15 = 0.55$.
(c) Such numbers are impossible: the intersection is part of A, so $P(A \cap B) \le P(A)$, but here $0.35 > 0.20$ (axiom 1 together with monotonicity is violated).
(d) Directly: there are 11 favourable pairs (six with a six on the first die and five on the second, not counting $(6, 6)$ again), $P = 11/36 = 0.306$. Via the complement: no sixes in $5 \cdot 5 = 25$ pairs, $P = 1 - 25/36 = 11/36 = 0.306$.

Problem 3 [B] Of 500 firms, 300 export, 200 import and 120 do both. Find the probabilities that a randomly chosen firm: (a) exports or imports; (b) does neither; (c) only exports; (d) only imports.

SOLUTION Let E be export, I import: $P(E) = 0.60$, $P(I) = 0.40$, $P(E \cap I) = 120/500 = 0.24$.

(a) $P(E \cup I) = 0.60 + 0.40 - 0.24 = 0.76$ (380 firms).
(b) $1 - 0.76 = 0.24$ (120 firms).
(c) Only export: $(300 - 120)/500 = 180/500 = 0.36$.
(d) Only import: $(200 - 120)/500 = 80/500 = 0.16$.

Check: $0.36 + 0.16 + 0.24 + 0.24 = 1$.

Problem 4 [B] Of 300 students, 180 study Finance and 120 study Marketing. In the retake exam 90 finance students and 40 marketing students passed.

(a) Build the contingency table.
(b) Find $P(\text{passed})$, $P(\text{passed} \mid \text{finance})$, $P(\text{passed} \mid \text{marketing})$.

(c) Check whether the programme and passing are independent.

SOLUTION

(a)

	Passed	Failed	Total
Finance	90	90	180
Marketing	40	80	120
Total	130	170	300

(b) $P(\text{passed}) = 130/300 = 0.433$; $P(\text{passed} \mid \text{fin}) = 90/180 = 0.50$; $P(\text{passed} \mid \text{mkt}) = 40/120 = 0.333$.

(c) If the events were independent, $P(\text{passed} \mid \text{fin})$ would equal $P(\text{passed})$. Here $0.50 \neq 0.433$. Equivalently: $P(\text{passed}) \cdot P(\text{fin}) = 0.433 \cdot 0.60 = 0.26$, but $P(\text{passed} \cap \text{fin}) = 90/300 = 0.30$. So the events are **dependent**.

Problem 5 [B] An analyst claims: "the default probability of each of three borrowers is 0.05, so the probability that at least one of them goes bankrupt is $3 \times 0.05 = 0.15$". Point out two errors in this reasoning and find the correct value assuming the borrowers are independent.

SOLUTION Error 1: the probabilities are added without accounting for intersections. The events "borrower 1 defaults", "borrower 2 defaults", "borrower 3 defaults" are not mutually exclusive (several defaults may occur at once), while the addition rule for mutually exclusive events applies only to mutually exclusive events. The sum overstates the probability.
Error 2: the independence assumption is not stated; it is what allows probabilities to be multiplied. In practice defaults are correlated in a crisis, and then a calculation under independence understates the risk.
Correct under independence: $P = 1 - 0.95^3 = 1 - 0.857 = 0.143$ (more precisely 0.1426).

Problem 6 [B] In how many ways can one:

(a) form a four-digit PIN code from distinct digits;
(b) choose a committee of 4 from 12 employees;
(c) choose a president, vice-president and treasurer from 10 analysts;
(d) rearrange the letters of the word "ECONOMICS"?

SOLUTION

(a) $10 \cdot 9 \cdot 8 \cdot 7 = 5\,040$ (arrangements, order matters).
(b) $\binom{12}{4} = 495$ (combinations, order does not matter).
(c) $P(10, 3) = 10 \cdot 9 \cdot 8 = 720$ (the posts are different).
(d) The word has 9 letters: O occurs 2 times, C 2 times. $\dfrac{9!}{2!\,2!} = \dfrac{362\,880}{4} = 90\,720$.

Problem 7 [B] A batch has 30 phones, 5 of them faulty. For inspection 3 are chosen at random (without replacement). Find the probabilities that among those inspected: (a) none is faulty; (b) exactly one is faulty; (c) at least one is faulty.

SOLUTION The total number of sets is $\binom{30}{3} = 4\,060$.
(a) $\binom{25}{3} = 2\,300$, $P = 2\,300/4\,060 = 0.567$.
(b) $\binom{5}{1}\binom{25}{2} = 5 \cdot 300 = 1\,500$, $P = 1\,500/4\,060 = 0.369$.
(c) $1 - 0.567 = 0.433$.

Problem 8 [B] (a) Find the probability that among 25 people there are at least two with a common birthday (assume the year has 365 days and all dates are equally likely). (b) Derive the formula and explain why it is easier to compute through the complement. (c) Find the answer for a group of 40 people.

SOLUTION (a), (b) The event "at least two coincide" is hard to break down by the number of coinciding pairs; the complement "all dates are different" is easy to compute: any date suits the first person, 364 out of 365 the second, 363 the third, etc.

$$P(\text{all different}) = \frac{365}{365} \cdot \frac{364}{365} \cdots \frac{341}{365}, \qquad P(\text{at least two coincide}) = 1 - P(\text{all different}).$$

For $n = 25$: $P = 0.569$.
(c) For $n = 40$: $P = 0.891$.

Problem 9 [C] A bank issued 600 loans: 360 to employees, of whom 18 stopped paying, and 240 to entrepreneurs, of whom 36 stopped paying.

(a) Build the contingency table "borrower category $\times$ default".
(b) Find $P(\text{default})$, $P(\text{default} \mid \text{entrepreneur})$, $P(\text{default} \mid \text{employee})$.
(c) Are borrower category and default independent?
(d) Find $P(\text{entrepreneur} \mid \text{default})$ and explain how this probability differs from $P(\text{default} \mid \text{entrepreneur})$.

SOLUTION

(a)

	Default	No default	Total
Employees	18	342	360
Entrepreneurs	36	204	240
Total	54	546	600

(b) $P(\text{default}) = 54/600 = 0.09$; $P(\text{default} \mid \text{entrep.}) = 36/240 = 0.15$; $P(\text{default} \mid \text{employee}) = 18/360 = 0.05$.
(c) No. $P(\text{default}) \cdot P(\text{entrep.}) = 0.09 \cdot 0.40 = 0.036$, while $P(\text{default} \cap \text{entrep.}) = 36/600 = 0.06$. Equivalently: $0.15 \neq 0.09$.
(d) $P(\text{entrep.} \mid \text{default}) = 36/54 = 0.667$. This is the share of entrepreneurs *among defaulters*, whereas 0.15 is the share of defaulters *among entrepreneurs*. The conditions are different, the denominators are different (54 and 240), so the numbers differ. Confusing $P(A \mid B)$ with $P(B \mid A)$ is one of the most common mistakes (in detail next week).

Problem 10 [C] For three events $P(A) = 0.50$, $P(B) = 0.40$, $P(C) = 0.30$, $P(A \cap B) = 0.20$, $P(A \cap C) = 0.15$, $P(B \cap C) = 0.10$, $P(A \cap B \cap C) = 0.05$.

(a) Find $P(A \cup B \cup C)$ by the inclusion–exclusion formula

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) \\ - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C).$$

(b) Find the probability that none of the three events occurs.
(c) Explain why the triple intersection is added at the end.

SOLUTION

(a) $0.50 + 0.40 + 0.30 - 0.20 - 0.15 - 0.10 + 0.05 = 0.80$.
(b) $1 - 0.80 = 0.20$.
(c) The sum $P(A) + P(B) + P(C)$ counts outcomes lying in exactly two sets twice, and those lying in all three, three times. Subtracting the three pairwise intersections removes one repeat from each "double" outcome, but outcomes from the triple intersection are also subtracted three times and end up not counted at all, so $P(A \cap B \cap C)$ must be added back once.

Problem 11 [C] It is known that $P(A) = 0.30$ and $P(B) = 0.50$.

(a) Let A and B be mutually exclusive. Find $P(A \cup B)$. Can they be independent?
(b) Let A and B be independent. Find $P(A \cap B)$ and $P(A \cup B)$.
(c) Can events be mutually exclusive and independent at the same time?

SOLUTION

(a) $P(A \cup B) = 0.30 + 0.50 = 0.80$. They cannot be independent: $P(A \cap B) = 0$, while $P(A)P(B) = 0.15 \neq 0$.
(b) $P(A \cap B) = 0.30 \cdot 0.50 = 0.15$; $P(A \cup B) = 0.30 + 0.50 - 0.15 = 0.65$.
(c) Only if the probability of at least one of them is zero: the conditions $P(A \cap B) = 0$ and $P(A \cap B) = P(A)P(B)$ together give $P(A)P(B) = 0$. For events with positive probability "mutually exclusive" and "independent" rule each other out.

Problem 12 [C] In the 17th century the Chevalier de Méré considered two bets equivalent. (a) What is the probability of rolling at least one six in 4 rolls of one die? (b) What is the probability of at least one "double six" in 24 rolls of two dice? Which bet is more favourable? (c) A coin has landed heads 5 times in a row. What is the probability of heads on the next toss, and why is it not > 0.5?

SOLUTION (a) $1 - (5/6)^4 = 1 - 0.482 = 0.518$.
(b) The probability of a "double six" per roll is $1/36$, the probability of none is $35/36$; $1 - (35/36)^{24} = 1 - 0.509 = 0.491$.
Bet (a) is favourable ($0.518 > 0.5$), bet (b) is unfavourable ($0.491 < 0.5$): de Méré was wrong to consider them equivalent.
(c) Exactly 0.5: tosses are independent, the coin does not "remember" earlier outcomes. The probability of *six* heads in a row before the tossing began was $1/64$, but after five heads the probability of the sixth is still $1/2$ (the gambler's fallacy).

ANSWERS AND SOLUTIONS · LECTURE 3

Conditional Probability and Bayes' Theorem

Problem 1 [A] It is known that $P(A \cap B) = 0.10$, $P(A) = 0.40$, $P(B) = 0.50$.

(a) Find $P(A \mid B)$ and $P(B \mid A)$.
(b) A student claims: "Since $P(A \mid B) = 0.8$, also $P(B \mid A) = 0.8$." Is he right?

SOLUTION

(a) $P(A \mid B) = \dfrac{0.10}{0.50} = 0.20$ (the denominator is the probability of the condition B);
$P(B \mid A) = \dfrac{0.10}{0.40} = 0.25$.

(b) No. The conditional probabilities $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$ and $P(B \mid A) = \frac{P(A \cap B)}{P(A)}$ have the same numerator but different denominators; they are equal only if $P(A) = P(B)$. In part (a) $0.20 \neq 0.25$.

Problem 2 [A] Of 1000 bank customers, 400 use the mobile app (M). Of these, 250 hold a deposit (D); among the other 600 customers, 120 hold a deposit.

(a) Find $P(\mathrm{D})$.
(b) Find $P(\mathrm{D} \mid \mathrm{M})$ and $P(\mathrm{D} \mid \overline{\mathrm{M}})$.
(c) Find $P(\mathrm{M} \mid \mathrm{D})$.

SOLUTION In total $250 + 120 = 370$ customers hold a deposit.

(a) $P(\mathrm{D}) = 370/1000 = 0.37$.
(b) $P(\mathrm{D} \mid \mathrm{M}) = 250/400 = 0.625$; $P(\mathrm{D} \mid \overline{\mathrm{M}}) = 120/600 = 0.20$.
(c) $P(\mathrm{M} \mid \mathrm{D}) = 250/370 = 0.676$.

Problem 3 [B] A factory makes parts on three machines A, B, C with output shares of 50%, 30%, 20%. The defect rates are 2%, 3%, 5% respectively.

(a) Find the probability that a random part is defective.
(b) A part turns out to be defective. Find the probability that it was made on machine C; on A; on B.
(c) Check that the three posterior probabilities sum to 1.

SOLUTION

(a) Law of total probability: $P(\text{defect}) = 0.5 \cdot 0.02 + 0.3 \cdot 0.03 + 0.2 \cdot 0.05 = 0.010 + 0.009 + 0.010 = 0.029$.

(b) $P(C \mid \text{defect}) = \dfrac{0.2 \cdot 0.05}{0.029} = \dfrac{0.010}{0.029} = 0.345$; $\quad P(A \mid \text{defect}) = \dfrac{0.010}{0.029} = 0.345$;
$P(B \mid \text{defect}) = \dfrac{0.009}{0.029} = 0.310$.

(c) $0.345 + 0.345 + 0.310 = 1$ (more precisely, $10/29 + 10/29 + 9/29 = 1$).

Remark: machine C makes only 20% of the parts but accounts for 34.5% of the defects.

Problem 4 [B] A bank catches fraudulent transactions. The share of fraudulent transactions is 2%. The system flags 85% of fraudulent transactions and wrongly flags 10% of normal ones.

(a) Find the probability that a flagged transaction is really fraudulent.
(b) Solve the same problem by the natural-frequency method on 10 000 transactions.
(c) Explain why the result is much smaller than the 85% sensitivity.

SOLUTION

(a) $P(F \mid \text{flag}) = \dfrac{0.85 \cdot 0.02}{0.85 \cdot 0.02 + 0.10 \cdot 0.98} = \dfrac{0.017}{0.017 + 0.098} = \dfrac{0.017}{0.115} = 0.148.$
(b) Of 10 000 transactions, 200 are fraudulent ($0.85 \cdot 200 = 170$ flagged) and 9800 are normal ($0.10 \cdot 9800 = 980$ flagged). In total 1150 are flagged, of which $170/1150 = 0.148$ are fraudulent.
(c) There are 49 times as many normal transactions as fraudulent ones, so even a small false-alarm rate (10%) gives 980 false flags against 170 correct ones. The sensitivity of 85% answers the question "what share of fraudulent transactions is flagged", but we need the reverse: "what share of flagged transactions are fraudulent".

Problem 5 [B] A disease occurs in 0.5% of the population. A test detects it in 98% of the sick and gives a false positive in 5% of the healthy.

(a) What is the probability of disease given a positive result?
(b) The test was repeated independently (given health status) and is again positive. What is the probability of disease now?

SOLUTION

(a) $P = \dfrac{0.98 \cdot 0.005}{0.98 \cdot 0.005 + 0.05 \cdot 0.995} = \dfrac{0.00490}{0.00490 + 0.04975} = 0.0897 \approx 9\%.$
(b) Prior odds $0.005/0.995 = 0.005025$; likelihood ratio $LR^{+} = 0.98/0.05 = 19.6$. After two positive tests the odds are $0.005025 \cdot 19.6^2 = 1.930$, hence $P = 1.930/2.930 = 0.659 \approx 66\%$.

Problem 6 [B] Continue the fraud problem (share 2%, sensitivity 85%, false alarms 10%). Suppose k independent (given fraud / no fraud) systems have raised a signal on the same transaction.

(a) Find the prior odds and the likelihood ratio LR^{+}.
(b) Find the probability of fraud for $k = 1,\ 2,\ 3$ signals.
(c) What happens to the result if the signals are not independent but fully duplicate each other?

SOLUTION

(a) Odds $0.02/0.98 = 0.0204$; $LR^{+} = 0.85/0.10 = 8.5$.
(b) The odds after k signals: $0.0204 \cdot 8.5^{k}$.
$k = 1$: 0.1735, $P = 0.1735/1.1735 = 0.148$;
$k = 2$: 1.475, $P = 1.475/2.475 = 0.596$;
$k = 3$: 12.54, $P = 12.54/13.54 = 0.926$.
(c) Duplicates carry no new information: the probability stays at 14.8% for any number of copies of the same signal. Likelihood ratios may be multiplied only for conditionally independent pieces of evidence.

Problem 7 [B] For a transaction in the fraud problem, the loss from a missed fraud is 8 million UZS, an investigation costs 0.4 million UZS (and completely eliminates the loss).

(a) Find the threshold probability above which an investigation pays off.
(b) Is it worthwhile to investigate a transaction that received one signal (fraud probability 14.8%)?
(c) At what minimum prior probability is one signal enough to justify an investigation?

SOLUTION

(a) We investigate if $p \cdot L > c$, i.e. $p > c/L = 0.4/8 = 0.05 = 5\%$.
(b) Yes: $14.8\% > 5\%$. The expected loss without investigation is $0.148 \cdot 8 = 1.18$ million, the investigation costs 0.4 million.
(c) We need posterior odds of $0.05/0.95 = 0.0526$, so the prior odds are $0.0526/8.5 = 0.00619$, and the prior probability is $0.00619/1.00619 = 0.00615 \approx 0.6\%$.

Problem 8 [B] For a transaction with posterior fraud probability $p = 0.148$, three actions are considered: (1) let it pass (expected loss: $8p$ million); (2) investigate (cost 0.4 million, no loss); (3) block (if the transaction is normal, the damage to the customer is 1 million; if fraudulent, no loss). Which action minimizes the expected loss?

SOLUTION Expected losses (million UZS):
let pass: $8 \cdot 0.148 = 1.183$;
investigate: 0.400;
block: $(1 - 0.148) \cdot 1 = 0.852$.
The minimum is for “investigate” (0.4 million).

Problem 9 [C]

(a) Derive Bayes’ formula for event A_i from the definition of conditional probability and the law of total probability (the events $A_1, \dots, A_n$ form a partition).
(b) Prove the odds form: $\dfrac{P(A \mid B)}{P(\overline{A} \mid B)} = \dfrac{P(A)}{P(\overline{A})} \cdot \dfrac{P(B \mid A)}{P(B \mid \overline{A})}$.

SOLUTION

(a) $P(A_i \mid B) = \dfrac{P(A_i \cap B)}{P(B)} = \dfrac{P(B \mid A_i)P(A_i)}{\sum_j P(B \mid A_j)P(A_j)}$: the numerator by the multiplication rule, the denominator by the law of total probability.
(b) By Bayes’ formula $P(A \mid B) = \frac{P(B|A)P(A)}{P(B)}$ and $P(\overline{A} \mid B) = \frac{P(B|\overline{A})P(\overline{A})}{P(B)}$. Dividing the first by the second, $P(B)$ cancels and the required equality remains.

Problem 10 [C] A family has two children.

(a) It is known that at least one child is a boy. What is the probability that both are boys?
(b) It is known that the older child is a boy. What is the probability that both are boys?

SOLUTION The sample space (older, younger): BB, BG, GB, GG, equally likely.

(a) The condition “at least one boy” leaves BB, BG, GB; one outcome is favourable: $1/3$.
(b) The condition “the older is a boy” leaves BB, BG: $1/2$.

> Different conditions — different answers.

Problem 11 [C] The Monty Hall problem. Behind one of three doors is a car, behind the other two are goats. You have chosen door 1. The host, who knows where the car is, opens a door with a goat (from the two remaining; if both have goats, he chooses at random) and offers you the chance to switch. Find the probability of winning if you switch.

SOLUTION If the car is behind door 1 (probability $1/3$), switching loses. If the car is behind door 2 or 3 (probability $2/3$), the host is forced to open the other door with a goat, and switching wins. In total $P(\text{win by switching}) = 2/3$, without switching $1/3$. Via Bayes: suppose you chose door 1 and the host opened door 3. $P(\text{opens } 3 \mid \text{car } 1) = 1/2$, $P(\text{opens } 3 \mid \text{car } 2) = 1$, $P(\text{opens } 3 \mid \text{car } 3) = 0$, so $P(\text{car } 2 \mid \text{opens } 3) = \dfrac{1 \cdot \frac{1}{3}}{\frac{1}{2} \cdot \frac{1}{3} + 1 \cdot \frac{1}{3}} = \dfrac{2}{3}$.

Problem 12 [C] Two fraud-alert systems (sensitivity 90%, false alarms 12%, fraud share 1%) raised signals. An analyst multiplied the likelihood ratios and obtained 36.2%. It then turned out that the second system simply copies the decision of the first. What should the estimate of the probability be?

SOLUTION The second system gives no independent information, so one signal remains: $P = \dfrac{0.9 \cdot 0.01}{0.9 \cdot 0.01 + 0.12 \cdot 0.99} = \dfrac{0.009}{0.1278} = 0.0704 = 7.04\%$. The analyst's error is applying independence where there is none: 36.2% overstates the probability about fivefold.

ANSWERS AND SOLUTIONS · LECTURE 4

Random Variables and Discrete Distributions

Problem 1 [A] The distribution of the variable X: $P(1) = 0.2$, $P(2) = 0.3$, $P(3) = k$, $P(4) = 0.1$.

(a) Find k.
(b) Find $\mathrm{E}[X]$, $\mathrm{Var}(X)$, $\mathrm{SD}(X)$.
(c) Find $P(X \geq 3)$.

SOLUTION

(a) $0.2 + 0.3 + k + 0.1 = 1 \Rightarrow k = 0.4$.
(b) $\mathrm{E}[X] = 0.2 + 0.6 + 1.2 + 0.4 = 2.4$; $\mathrm{E}[X^2] = 0.2 + 1.2 + 3.6 + 1.6 = 6.6$; $\mathrm{Var}(X) = 6.6 - 2.4^2 = 0.84$; $\mathrm{SD}(X) = 0.917$.
(c) $P(X \geq 3) = 0.4 + 0.1 = 0.5$.

Problem 2 [A] For a variable X: $\mathrm{E}[X] = 40$, $\mathrm{SD}(X) = 6$. The variable X_2 is independent of X: $\mathrm{E}[X_2] = 25$, $\mathrm{SD}(X_2) = 8$.

(a) Find the expectation, variance and standard deviation of $Y = 3X - 10$.
(b) Find the expectation and standard deviation of $X + X_2$ and $X - X_2$.

SOLUTION

(a) $\mathrm{E}[Y] = 3 \cdot 40 - 10 = 110$; $\mathrm{Var}(Y) = 3^2 \cdot 36 = 324$; $\mathrm{SD}(Y) = 18$.
(b) Expectations: $40 + 25 = 65$ and $40 - 25 = 15$. The variance in both cases is $36 + 64 = 100$, so $\mathrm{SD} = 10$ (subtraction does not reduce the spread).

Problem 3 [A] $X \sim \mathrm{Bin}(5, 0.4)$.

(a) Build the table of $P(X = k)$, $k = 0, \ldots, 5$.
(b) Find $\mathrm{E}[X]$ and $\mathrm{Var}(X)$ by the formulas.
(c) Find $P(X \geq 4)$.

SOLUTION

(a) $P(k) = \binom{5}{k} 0.4^k 0.6^{5-k}$:

k	0	1	2	3	4	5
P	0.0778	0.2592	0.3456	0.2304	0.0768	0.0102

(b) $\mathrm{E}[X] = 5 \cdot 0.4 = 2$; $\mathrm{Var}(X) = 5 \cdot 0.4 \cdot 0.6 = 1.2$.
(c) $P(X \geq 4) = 0.0768 + 0.0102 = 0.0870$.

Problem 4 [B] A borrower repays a loan with probability 0.9 independently of the others. The bank issued 12 loans.

(a) Find the expected number of repaid loans and the standard deviation.

(b) Find the probability that all 12 are repaid.
(c) Find the probability that at most one loan is not repaid.

SOLUTION $X \sim \text{Bin}(12, 0.9)$.
(a) $\text{E}[X] = 10.8$, $\sigma = \sqrt{12 \cdot 0.9 \cdot 0.1} = \sqrt{1.08} = 1.04$.
(b) $P(X = 12) = 0.9^{12} = 0.2824$.
(c) "At most one non-repayment" means $X \geq 11$: $P(X = 11) = 12 \cdot 0.9^{11} \cdot 0.1 = 0.3766$, so $P(X \geq 11) = 0.3766 + 0.2824 = 0.659$.

Problem 5 [B] The average number of complaints to a support line is 1.5 per hour (a Poisson flow).

(a) Find the probability that there are no complaints in an hour.
(b) Find the probability of at least two complaints in an hour.
(c) Find the probability that at most one complaint arrives in two hours.

SOLUTION
(a) $P(0) = e^{-1.5} = 0.2231$.
(b) $P(X \geq 2) = 1 - P(0) - P(1) = 1 - 0.2231 - 1.5 \cdot 0.2231 = 1 - 0.2231 - 0.3347 = 0.4422$.
(c) For two hours $\lambda = 3$: $P(X \leq 1) = e^{-3}(1 + 3) = 0.0498 \cdot 4 = 0.1991$.

Problem 6 [B] Two projects are considered. Project A: profit $+10$ million with probability 0.6 and -5 million with probability 0.4. Project B: $+6$ million with probability 0.7 and -1 million with probability 0.3.

(a) Find the expected profit and standard deviation of each project.
(b) Which project would you choose if the goal is to maximize expected profit? And if stability matters? Justify.

SOLUTION
(a) A: $\text{E}[X] = 6 - 2 = 4$; $\text{E}[X^2] = 60 + 10 = 70$; Var $= 70 - 16 = 54$; SD $= 7.35$.
B: $\text{E}[X] = 4.2 - 0.3 = 3.9$; $\text{E}[X^2] = 25.2 + 0.3 = 25.5$; Var $= 25.5 - 15.21 = 10.29$; SD $= 3.21$.
(b) By expectation A is slightly better (4.0 against 3.9), but the spread of A is more than twice as large (7.35 against 3.21), and the probability of a loss is 40% against 30%. A gain of 0.1 million does not justify such an increase in risk, so with a cautious policy B is sensible. (Spread per unit of expected profit: $7.35/4 = 1.84$ against $3.21/3.9 = 0.82$.)

Problem 7 [B] An insurance company sells a policy for 80 thousand UZS. Annual payouts on the policy: 0 with probability 0.94, 400 thousand with probability 0.05, 3000 thousand with probability 0.01.

(a) Find the expected payout and the standard deviation of the payout on one policy.
(b) Find the expected profit and the standard deviation of profit on one policy.
(c) The company sold 1000 independent policies. Find the expected total profit and its standard deviation.

SOLUTION

(a) $\mathrm{E}[Y] = 400 \cdot 0.05 + 3000 \cdot 0.01 = 20 + 30 = 50$; $\mathrm{E}[Y^2] = 160000 \cdot 0.05 + 9\,000\,000 \cdot 0.01 = 8000 + 90\,000 = 98\,000$; $\mathrm{Var}(Y) = 98\,000 - 2500 = 95\,500$; $\mathrm{SD}(Y) = 309.0$ (thousand UZS).

(b) Profit $\Pi = 80 - Y$: $\mathrm{E}[\Pi] = 80 - 50 = 30$; $\mathrm{SD}(\Pi) = \mathrm{SD}(Y) = 309.0$ (a shift does not change the spread).

(c) $\mathrm{E}[T] = 1000 \cdot 30 = 30\,000$; $\mathrm{Var}(T) = 1000 \cdot 95\,500$, $\mathrm{SD}(T) = \sqrt{95\,500\,000} = 9772$ thousand UZS. The relative risk falls: $309/30 \approx 10$ per policy against $9772/30\,000 \approx 0.33$ for the portfolio.

Problem 8 [B] The profit of one loan has $\mu = 0.62$ million, $\sigma = 2.0$ million (Lecture 4, the loan example). The bank issues 25 independent loans. Find the expectation and standard deviation of the total profit and of the average profit per loan. How many times less variable is the average than one loan?

SOLUTION $\mathrm{E}[T] = 25 \cdot 0.62 = 15.5$; $\mathrm{Var}(T) = 25 \cdot 3.9916 = 99.79$, $\mathrm{SD}(T) = 9.99 \approx 10.0$. Average profit: $\mathrm{E}\big[\bar{X}\big] = 0.62$, $\mathrm{SD}(\bar{X}) = 9.99/25 = 0.40 = 2.0/\sqrt{25}$. The spread of the average is smaller by a factor of $\sqrt{25} = 5$.

Problem 9 [B] A bank processes 300 transactions per minute, each erroneous with probability 0.006 independently.

(a) Find the probability of no errors by the exact binomial formula and by the Poisson approximation.

(b) Find the probability of at least three errors by both methods.

SOLUTION $X \sim \mathrm{Bin}(300, 0.006)$, $\lambda = np = 1.8$.

(a) Exactly: $0.994^{300} = 0.1644$; Poisson: $e^{-1.8} = 0.1653$.

(b) Poisson: $P(X \ge 3) = 1 - e^{-1.8}(1 + 1.8 + 1.62) = 1 - 0.1653 \cdot 4.42 = 0.2694$; binomial — 0.2692. The approximation differs in the third decimal place.

Problem 10 [B] The probability of closing a deal on one call is 0.25 (calls are independent). A manager calls until the first deal.

(a) What is the probability that the first deal comes on the fourth call?

(b) What is the probability that at most four calls suffice?

(c) How many calls are needed on average?

SOLUTION

(a) $0.75^3 \cdot 0.25 = 0.1055$.

(b) $1 - 0.75^4 = 1 - 0.3164 = 0.6836$.

(c) $1/p = 1/0.25 = 4$.

Problem 11 [C] Prove: (a) $\mathrm{Var}(X) = \mathrm{E}\big[X^2\big] - \mu^2$; (b) $\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$.

SOLUTION (a) $\mathrm{E}\big[(X - \mu)^2\big] = \mathrm{E}\big[X^2 - 2\mu X + \mu^2\big] = \mathrm{E}\big[X^2\big] - 2\mu \cdot \mu + \mu^2 = \mathrm{E}\big[X^2\big] - \mu^2$ (linearity of expectation).

(b) $\mathrm{E}[aX+b] = a\mu + b$, so $aX + b - \mathrm{E}[aX+b] = a(X-\mu)$ and $\mathrm{Var}(aX+b) = \mathrm{E}\big[a^2(X-\mu)^2\big] = a^2\,\mathrm{Var}(X)$. The constant b disappears upon subtracting the mean.

Problem 12 [C] Let $X \sim \mathrm{Bin}(n, p)$ be the number of successes, $X = I_1 + \cdots + I_n$, where I_i is the indicator of success in the i-th trial ($I_i = 1$ with probability p, otherwise 0, independent).

(a) Find $\mathrm{E}[I_i]$ and $\mathrm{Var}(I_i)$.
(b) Derive $\mathrm{E}[X] = np$ and $\mathrm{Var}(X) = np(1-p)$.
(c) At what p is the variance of a single trial maximal? What is it equal to?

SOLUTION

(a) $\mathrm{E}[I_i] = p$; $\mathrm{E}\big[I_i^2\big] = p$ (since $I_i^2 = I_i$); $\mathrm{Var}(I_i) = p - p^2 = p(1-p)$.
(b) The expectation of a sum is the sum of expectations: np. For independent summands the variances add: $np(1-p)$.
(c) $p(1-p)$ is maximal at $p = 0.5$ and equals 0.25 (a parabola with its vertex at $p = 1/2$). A fair coin has the greatest uncertainty.

ANSWERS AND SOLUTIONS · LECTURE 5

Continuous Distributions and the Normal Distribution

Problem 1 [A] A bus runs at 15-minute intervals. A passenger arrives at a random moment, the waiting time is $X \sim U(0, 15)$.

(a) Find μ and σ.
(b) Find $P(X > 10)$ and $P(5 < X < 8)$.

SOLUTION

(a) $\mu = 7.5$ min, $\sigma = 15/\sqrt{12} = 4.33$ min.
(b) $P(X > 10) = 5/15 = 1/3 = 0.333$; $P(5 < X < 8) = 3/15 = 0.2$.

Problem 2 [A] The service time of a customer at a bank is exponential with a mean of 5 minutes.

(a) What is the parameter λ?
(b) Find the probability that service lasts more than 3 minutes, and the probability that it lasts less than 10 minutes.
(c) Find the median.

SOLUTION

(a) $\lambda = 1/5 = 0.2$.
(b) $P(X > 3) = e^{-0.6} = 0.5488$; $P(X < 10) = 1 - e^{-2} = 0.8647$.
(c) $0.5 = e^{-0.2m} \Rightarrow m = 5 \ln 2 = 3.47$ min (less than the mean).

Problem 3 [A] $X \sim N(200, 20^2)$. Find $P(X < 230)$, $P(X > 170)$, $P(190 < X < 215)$.

SOLUTION $z = (230 - 200)/20 = 1.5$: $P(X < 230) = \Phi(1.5) = 0.9332$.
$z = -1.5$: $P(X > 170) = 1 - \Phi(-1.5) = \Phi(1.5) = 0.9332$.
z for 190 and 215: -0.5 and 0.75: $P = \Phi(0.75) - \Phi(-0.5) = 0.7734 - 0.3085 = 0.4649$.

Problem 4 [B] A machine packs coffee, the mass of a pack is $X \sim N(500, 4^2)$ g.

(a) Find the share of packs weighing less than 495 g.
(b) Find the share of packs weighing more than 510 g.
(c) To what mean should the machine be set so that only 1% of packs weigh less than 495 g (the standard deviation stays the same)?

SOLUTION

(a) $z = (495 - 500)/4 = -1.25$, $P = 0.1056$ (10.6%).
(b) $z = 2.5$, $P = 1 - \Phi(2.5) = 1 - 0.9938 = 0.0062$.
(c) We need $\dfrac{495 - \mu}{4} = -2.3263$, hence $\mu = 495 + 9.31 = 504.3$ g.

Problem 5 [B] The size of a consumer loan is $X \sim N(40, 12^2)$ million UZS.

(a) Find the share of loans larger than 60 million.
(b) Find the share of loans smaller than 30 million.
(c) Find the cut-off for the largest 5% of loans and the interval containing the central 90% of loans.

SOLUTION
(a) $z = 20/12 = 1.667$, $P = 1 - \Phi(1.67) = 1 - 0.9525 = 0.0478$.
(b) $z = -10/12 = -0.833$, $P = \Phi(-0.83) = 0.2023$.
(c) Top 5%: $x = 40 + 1.6449 \cdot 12 = 59.7$ million. Central 90%: $40 \pm 1.6449 \cdot 12$, i.e. $(20.3;\ 59.7)$ million.

Problem 6 [B] For $X \sim N(90, \sigma^2)$ it is known that $P(X > 100) = 0.10$. For $Y \sim N(\mu, 8^2)$ it is known that $P(Y < 50) = 0.20$. Find σ and μ.

SOLUTION $z_{0.90} = 1.2816$: $\dfrac{100 - 90}{\sigma} = 1.2816$, $\sigma = 7.80$.
$z_{0.20} = -0.8416$: $\dfrac{50 - \mu}{8} = -0.8416$, $\mu = 50 + 6.73 = 56.73$.

Problem 7 [B] Student A scored 82 on an exam $N(70, 8^2)$, student B scored 640 on an exam $N(580, 60^2)$. Who showed the better relative result? What share of test-takers scored lower than each?

SOLUTION $z_A = (82 - 70)/8 = 1.5$, $z_B = (640 - 580)/60 = 1.0$. A is better. The share who scored lower: $\Phi(1.5) = 93.3\%$ against $\Phi(1.0) = 84.1\%$.

Problem 8 [B] The returns of two independent assets (%): $R_1 \sim N(8, 4^2)$, $R_2 \sim N(5, 3^2)$. The portfolio is the sum $R_1 + R_2$.

(a) Find the distribution of $R_1 + R_2$.
(b) Find the probability of a negative total return.
(c) Find the probability that $R_1 < R_2$.

SOLUTION
(a) $\mu = 13$, $\sigma^2 = 16 + 9 = 25$, $\sigma = 5$: $R_1 + R_2 \sim N(13, 5^2)$.
(b) $z = (0 - 13)/5 = -2.6$: $P = 0.0047$.
(c) $R_1 - R_2 \sim N(3, 5^2)$: $P(R_1 - R_2 < 0) = \Phi(-0.6) = 0.2743$.

Problem 9 [B] 40% of a shop's visitors make a purchase, independently. 200 visitors came in a day.

(a) Check that the normal approximation is applicable and find μ and σ of the number of buyers.
(b) Find the probability that there are at least 90 buyers (with a continuity correction).

SOLUTION $X \sim \text{Bin}(200, 0.4)$.
(a) $np = 80 \geq 10$ and $n(1-p) = 120 \geq 10$ — the approximation is admissible. $\mu = 80$, $\sigma = \sqrt{48} = 6.93$.
(b) $P(X \geq 90) = P(X > 89.5)$ (the discrete "at least 90" starts at 89.5): $z = (89.5 - 80)/6.93 = 1.37$, $P = 1 - \Phi(1.37) = 0.0853$. The exact binomial value is 0.0857; without

the correction we would get 0.0970.

Problem 10 [B] The time between customer arrivals is exponential with a mean of 2 minutes.

(a) Find the probability of waiting longer than 3 minutes for the next customer.
(b) Two minutes have already passed with no customers. What is the probability that we must wait at least 3 more minutes?
(c) What is the distribution of the number of customers in 10 minutes? Find the probability that nobody arrives in 10 minutes.

SOLUTION $\lambda = 0.5$ customers per minute.
(a) $e^{-0.5\cdot 3} = e^{-1.5} = 0.2231$.
(b) Memorylessness: the same, 0.2231.
(c) The number of customers $\sim$ Pois(5); $P(0) = e^{-5} = 0.0067$ (the same as $P(X > 10) = e^{-0.5\cdot 10}$).

Problem 11 [C] Prove the memorylessness of the exponential distribution: $P(X > s + t \mid X > s) = P(X > t)$.

SOLUTION The event $\{X > s + t\}$ is contained in $\{X > s\}$, so $P(X > s + t \mid X > s) = \dfrac{P(X > s + t)}{P(X > s)} = \dfrac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t)$.

Problem 12 [C] For $X \sim U(a, b)$ derive $\mathrm{E}[X]$ and $\mathrm{Var}(X)$ by integration.

SOLUTION $f = 1/(b-a)$ on $[a, b]$. $\mathrm{E}[X] = \frac{1}{b-a}\int_a^b x\,dx = \frac{b^2-a^2}{2(b-a)} = \frac{a+b}{2}$. $\mathrm{E}[X^2] = \frac{1}{b-a}\cdot \frac{b^3-a^3}{3} = \frac{a^2+ab+b^2}{3}$. $\mathrm{Var} = \frac{a^2+ab+b^2}{3} - \frac{(a+b)^2}{4} = \frac{4a^2+4ab+4b^2-3a^2-6ab-3b^2}{12} = \frac{(b-a)^2}{12}$.

Problem 13 [C] Let $X \sim N(\mu, \sigma^2)$. Show that the z-score $Z = (X - \mu)/\sigma$ has $\mathrm{E}[Z] = 0$ and $\mathrm{Var}(Z) = 1$. Explain why $IQR \approx 1.349\sigma$ implies that for $IQR = 27$ in normal data $\sigma \approx 20$.

SOLUTION $Z = \frac{1}{\sigma}X - \frac{\mu}{\sigma}$, so $\mathrm{E}[Z] = \frac{\mu}{\sigma} - \frac{\mu}{\sigma} = 0$ and $\mathrm{Var}(Z) = \frac{1}{\sigma^2}\sigma^2 = 1$ (properties of linear transformations, Lecture 4). The quartiles of the standard normal are ± 0.6745, so $IQR = 2 \cdot 0.6745\sigma = 1.349\sigma$ and $\sigma \approx 27/1.349 = 20.0$.

ANSWERS AND SOLUTIONS · LECTURE 6

Joint Distributions, Covariance and Correlation

Problem 1 [A] The joint distribution of two indicators: $P(X=0,Y=0)=0.3$, $P(X=0,Y=1)=0.2$, $P(X=1,Y=0)=0.1$, $P(X=1,Y=1)=0.4$.

(a) Find the marginal distributions of X and Y.
(b) Find $\mathrm{Cov}(X,Y)$ and ρ.
(c) Find $P(Y=1 \mid X=1)$. Are X and Y independent?

SOLUTION

(a) $P(X=0)=0.5$, $P(X=1)=0.5$; $P(Y=0)=0.4$, $P(Y=1)=0.6$.
(b) $\mathrm{E}[X]=0.5$, $\mathrm{E}[Y]=0.6$, $\mathrm{E}[XY]=P(X=1,Y=1)=0.4$; $\mathrm{Cov}=0.4-0.3=0.1$. $\mathrm{Var}(X)=0.25$, $\mathrm{Var}(Y)=0.24$; $\rho=0.1/\sqrt{0.06}=0.408$.
(c) $P(Y=1 \mid X=1)=0.4/0.5=0.8 \neq P(Y=1)=0.6$, so the variables are dependent.

Problem 2 [A] $\mathrm{Var}(X)=9$, $\mathrm{Var}(Y)=16$, $\mathrm{Cov}(X,Y)=6$. Find ρ, $\mathrm{Var}(X+Y)$, $\mathrm{Var}(X-Y)$, $\mathrm{Var}(2X-3Y)$.

SOLUTION $\rho=6/(3\cdot 4)=0.5$. $\mathrm{Var}(X+Y)=9+16+2\cdot 6=37$. $\mathrm{Var}(X-Y)=9+16-12=13$. $\mathrm{Var}(2X-3Y)=4\cdot 9+9\cdot 16-2\cdot 2\cdot 3\cdot 6=36+144-72=108$.

Problem 3 [A] Check independence in two tables (rows $Y=0,1$; columns $X=0,1$):
(1) $p(0,0)=0.2$, $p(1,0)=0.3$, $p(0,1)=0.2$, $p(1,1)=0.3$;
(2) $p(0,0)=0.1$, $p(1,0)=0.4$, $p(0,1)=0.3$, $p(1,1)=0.2$.
(The first index is the value of X, the second the value of Y.)

SOLUTION (1) $P(X=0)=0.4$, $P(X=1)=0.6$, $P(Y=0)=P(Y=1)=0.5$. All the products $0.4\cdot 0.5=0.2$, $0.6\cdot 0.5=0.3$ match the cells — *independent*.
(2) $P(X=0)=0.4$, $P(Y=0)=0.5$, but $p(0,0)=0.1\neq 0.4\cdot 0.5=0.2$ — *dependent*.

Problem 4 [B] Five students: study time x (h) $=1,2,3,4,5$ and score $y=50,55,65,70,85$.

(a) Find $\bar{x}$, $\bar{y}$ and $\sum(x_i-\bar{x})(y_i-\bar{y})$.
(b) Find the sample covariance and correlation.

SOLUTION

(a) $\bar{x}=3$, $\bar{y}=65$. Products of deviations: $(-2)(-15)=30$, $(-1)(-10)=10$, 0, $1\cdot 5=5$, $2\cdot 20=40$; sum 85.
(b) $s_{xy}=85/4=21.25$. $\sum(x_i-\bar{x})^2=10$, $\sum(y_i-\bar{y})^2=750$ $(225+100+0+25+400)$. $r=85/\sqrt{10\cdot 750}=0.981$.

Problem 5 [B] Assets: $\mu_1=9\%$, $\sigma_1=12\%$; $\mu_2=14\%$, $\sigma_2=20\%$; $\rho=0.2$. Portfolio: 40% in asset 1 and 60% in asset 2.

(a) Find the expected return and standard deviation of the portfolio.
(b) Compare σ_p with the weighted average of the standard deviations. What does the difference show?

SOLUTION

(a) $\mathrm{E}[R_p] = 0.4 \cdot 9 + 0.6 \cdot 14 = 12\%$. $\sigma_p^2 = 0.16 \cdot 144 + 0.36 \cdot 400 + 2 \cdot 0.4 \cdot 0.6 \cdot 0.2 \cdot 12 \cdot 20 = 23.04 + 144 + 23.04 = 190.08$, $\sigma_p = 13.79\%$.
(b) The weighted average: $0.4 \cdot 12 + 0.6 \cdot 20 = 16.8\%$. The portfolio is less risky (13.8%) — the diversification effect at $\rho < 1$.

Problem 6 [B] $\sigma_1 = 10\%$, $\sigma_2 = 15\%$, the assets are uncorrelated ($\rho = 0$). At what weight w of asset 1 is the portfolio variance minimal? What is σ_p?

SOLUTION For $\rho = 0$: $w^* = \dfrac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2} = \dfrac{225}{325} = 0.692$. Then $\sigma_p^2 = 0.692^2 \cdot 100 + 0.308^2 \cdot 225 = 47.9 + 21.3 = 69.2$, $\sigma_p = 8.32\%$ — less than either asset alone.

Problem 7 [B] For the data of Problem 4, study time was measured not in hours but in minutes ($x' = 60x$), and scores as a fraction of the maximum (suppose the maximum is 100, $y' = y/100$). How do s_{xy} and r change? Compute.

SOLUTION $s_{x'y'} = 60 \cdot \frac{1}{100} \cdot s_{xy} = 0.6 \cdot 21.25 = 12.75$ (covariance depends on units). The correlation does not change: $r = 0.981$, since scaling with positive factors preserves the sign and the ratio.

Problem 8 [B] $X \in \{0, 1, 2\}$ is the number of complaints per week, $Y \in \{1, 2\}$ the shift. Joint probabilities $p(x, y)$: $p(0,1) = 0.1$, $p(1,1) = 0.2$, $p(2,1) = 0.1$, $p(0,2) = 0.2$, $p(1,2) = 0.3$, $p(2,2) = 0.1$.

(a) Find the marginal distributions and $\mathrm{E}[X]$, $\mathrm{E}[Y]$.
(b) Find $\mathrm{Cov}(X, Y)$ and ρ.
(c) Find the conditional distribution of X given $Y = 2$ and $\mathrm{E}[X \mid Y = 2]$.

SOLUTION

(a) $P(X = 0) = 0.3$, $P(X = 1) = 0.5$, $P(X = 2) = 0.2$; $P(Y = 1) = 0.4$, $P(Y = 2) = 0.6$. $\mathrm{E}[X] = 0.5 + 0.4 = 0.9$, $\mathrm{E}[Y] = 0.4 + 1.2 = 1.6$.
(b) $\mathrm{E}[XY] = 1 \cdot 1 \cdot 0.2 + 2 \cdot 1 \cdot 0.1 + 1 \cdot 2 \cdot 0.3 + 2 \cdot 2 \cdot 0.1 = 0.2 + 0.2 + 0.6 + 0.4 = 1.4$. $\mathrm{Cov} = 1.4 - 0.9 \cdot 1.6 = -0.04$. $\mathrm{Var}(X) = 1.3 - 0.81 = 0.49$, $\mathrm{Var}(Y) = 2.8 - 2.56 = 0.24$, $\rho = -0.04/\sqrt{0.49 \cdot 0.24} = -0.117$.
(c) $P(X = 0 \mid Y = 2) = 0.2/0.6 = 0.333$, $P(X = 1 \mid Y = 2) = 0.5$, $P(X = 2 \mid Y = 2) = 0.167$; $\mathrm{E}[X \mid Y = 2] = 0.5 + 0.333 = 0.833$ (against $\mathrm{E}[X] = 0.9$).

Problem 9 [B] The costs of two components: $\sigma_1 = 4$, $\sigma_2 = 6$. Find the standard deviation of the total cost for $\rho = 0.5$, $\rho = 0$ and $\rho = -0.5$.

SOLUTION $\mathrm{Var} = 16 + 36 + 2\rho \cdot 24$: for $\rho = 0.5$: 76, $\sigma = 8.72$; for $\rho = 0$: 52, $\sigma = 7.21$; for $\rho = -0.5$: 28, $\sigma = 5.29$. A positive relationship increases the variability of the sum.

Problem 10 [B] A bank issued 50 loans, each with $\sigma = 2.0$ million. Find the standard deviation of the average profit for pairwise correlation $\rho = 0$ and $\rho = 0.2$; find the limit as $n \to \infty$ for $\rho = 0.2$.

SOLUTION For $\rho = 0$: $2/\sqrt{50} = 0.283$. For $\rho = 0.2$: $2\sqrt{(1 + 49 \cdot 0.2)/50} = 2\sqrt{10.8/50} = 0.930$. Limit: $\sigma\sqrt{\rho} = 2\sqrt{0.2} = 0.894$. Correlation creates a "floor" of risk that does not decrease as the number of loans grows.

Problem 11 [C] Prove that (a) $\mathrm{Cov}(X, Y) = \mathrm{E}[XY] - \mu_X\mu_Y$; (b) $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X, Y)$.

SOLUTION (a) $\mathrm{E}[(X - \mu_X)(Y - \mu_Y)] = \mathrm{E}[XY] - \mu_Y\mathrm{E}[X] - \mu_X\mathrm{E}[Y] + \mu_X\mu_Y = \mathrm{E}[XY] - \mu_X\mu_Y$.
(b) $\mathrm{Var}(X + Y) = \mathrm{E}\big[((X - \mu_X) + (Y - \mu_Y))^2\big] = \mathrm{E}\big[(X - \mu_X)^2\big] + \mathrm{E}\big[(Y - \mu_Y)^2\big] + 2\mathrm{E}[(X - \mu_X)(Y - \mu_Y)]$.

Problem 12 [C] Let $Y = aX + b$, $a \neq 0$. Show that $\rho_{XY} = +1$ for $a > 0$ and -1 for $a < 0$.

SOLUTION $\mathrm{Cov}(X, aX + b) = a\,\mathrm{Var}(X)$, $\sigma_Y = |a|\sigma_X$. Then $\rho = \dfrac{a\sigma_X^2}{\sigma_X \cdot |a|\sigma_X} = \dfrac{a}{|a|} = \pm 1$. The absolute value of the correlation equals 1 only for a strictly linear dependence.

Problem 13 [C] X takes the values $-1, 0, 1$ with probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$, and $Y = X^2$. Find $\mathrm{Cov}(X, Y)$ and prove that X and Y are dependent.

SOLUTION $\mathrm{E}[X] = 0$; $\mathrm{E}[XY] = \mathrm{E}[X^3] = (-1)\frac{1}{4} + 0 + 1 \cdot \frac{1}{4} = 0$; $\mathrm{Cov} = 0 - 0 \cdot \mathrm{E}[Y] = 0$.
But $P(X = 0, Y = 0) = \frac{1}{2} \neq P(X = 0)P(Y = 0) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$. A zero covariance does not guarantee independence.

ANSWERS AND SOLUTIONS · LECTURE 7

Sampling Distributions and the Central Limit Theorem

Problem 1 [A] The average bill in a chain of cafés is $\mu = 50$ thousand UZS, $\sigma = 12$. A sample of $n = 36$ bills is taken.

(a) Find the expectation and standard error of $\bar{X}$.
(b) Find $P(\bar{X} > 53)$ and $P(48 < \bar{X} < 52)$.

SOLUTION

(a) $\mathrm{E}[\bar{X}] = 50$, $\mathrm{SE} = 12/\sqrt{36} = 2$.
(b) $P(\bar{X} > 53) = P(Z > 1.5) = 0.0668$; $P(48 < \bar{X} < 52) = P(-1 < Z < 1) = 0.6827$.

Problem 2 [A] $\sigma = 20$. Find the standard error of the mean for $n = 25$ and $n = 100$. What sample size is needed for $\mathrm{SE} = 1$?

SOLUTION $20/5 = 4$; $20/10 = 2$. For $\mathrm{SE} = 1$: $\sqrt{n} = 20$, $n = 400$ (to reduce the error fourfold, 16 times as many observations are needed).

Problem 3 [A] The share of customers with the mobile app is $p = 0.30$. $n = 200$ were surveyed. Find $\mathrm{SE}(\hat{p})$, $P(\hat{p} > 0.35)$ and $P(0.28 < \hat{p} < 0.32)$.

SOLUTION $\mathrm{SE} = \sqrt{0.3 \cdot 0.7/200} = 0.0324$. $z = 0.05/0.0324 = 1.54$: $P(\hat{p} > 0.35) = 0.0614$. $z = \pm 0.02/0.0324 = \pm 0.62$: $P = 2\Phi(0.62) - 1 \approx 0.465$ (more precisely, 0.463).

Problem 4 [B] The size of a deposit is strongly right-skewed: $\mu = 8$ million, $\sigma = 5$ million. A sample of $n = 50$ deposits.

(a) May the normal distribution be used for $\bar{X}$? Justify.
(b) Find $P(\bar{X} < 7)$ and $P(\bar{X} > 9.5)$.
(c) Can $P(X < 7)$ be found in the same way for a single deposit?

SOLUTION

(a) Yes: by the CLT, for $n = 50 \geq 30$ the distribution of $\bar{X}$ is approximately normal, although the original one is skewed.
(b) $\mathrm{SE} = 5/\sqrt{50} = 0.707$. $P(\bar{X} < 7) = P(Z < -1.41) = 0.0786$; $P(\bar{X} > 9.5) = P(Z > 2.12) = 0.0170$.
(c) No: for a single observation the CLT does not apply, and the distribution is skewed — a normal approximation is inadmissible.

Problem 5 [B] Each of 100 transactions has $\mu = 2$ million UZS and $\sigma = 1.5$ million, the transactions are independent. Find the approximate probability that the total volume exceeds 230 million.

SOLUTION $T = \sum X_i \approx N(100 \cdot 2,\ 100 \cdot 1.5^2) = N(200, 15^2)$. $z = (230 - 200)/15 = 2$, $P = 1 - \Phi(2) = 0.0228$.

Problem 6 [B] We need to estimate the average time to resolve a request ($\sigma \approx 15$ minutes) to within $E = 2$ minutes at 95% confidence; then to estimate the share of satisfied customers to within ± 0.04.

(a) Find the required n for the mean.
(b) How does n change if the required precision is $E = 1$ minute?
(c) Find n for the proportion with unknown p and with $p \approx 0.2$.

SOLUTION

(a) $n \geq (1.96 \cdot 15/2)^2 = 216.1$, i.e. $n = 217$.
(b) $n \geq (1.96 \cdot 15)^2 = 864.4$, i.e. 865 — four times as many for half the error.
(c) Unknown p ($p = 0.5$): $n \geq 1.96^2 \cdot 0.25/0.04^2 = 600.25$, i.e. 601. For $p \approx 0.2$: $n \geq 1.96^2 \cdot 0.16/0.04^2 = 384.2$, i.e. 385.

Problem 7 [B] The defect rate on a line is $p = 0.05$. A batch of $n = 400$ parts is inspected.

(a) Check the conditions of the normal approximation and find $\mathrm{SE}(\hat{p})$.
(b) Find the probability that the sample defect rate exceeds 7%.
(c) Find the probability that the sample defect rate turns out below 3%.

SOLUTION

(a) $np = 20 \geq 10$, $n(1-p) = 380 \geq 10$. $\mathrm{SE} = \sqrt{0.05 \cdot 0.95/400} = 0.0109$.
(b) $z = 0.02/0.0109 = 1.84$; $P(\hat{p} > 0.07) = 0.0332$.
(c) By symmetry $P(\hat{p} < 0.03) = 0.0332$.

Problem 8 [B] The population is $\{2, 4, 6\}$; samples of 2 with replacement are taken (9 equally likely).

(a) Find μ and σ^2.
(b) Write the distribution of $\bar{X}$ and check $\mathrm{E}[\bar{X}] = \mu$, $\mathrm{Var}(\bar{X}) = \sigma^2/2$.
(c) Find the mean value of s^2 (with divisor $n - 1$).

SOLUTION

(a) $\mu = 4$; $\sigma^2 = \frac{4+0+4}{3} = 2.667$.
(b) $\bar{x} = 2, 3, 4, 5, 6$ with probabilities $\frac{1}{9}, \frac{2}{9}, \frac{3}{9}, \frac{2}{9}, \frac{1}{9}$. $\mathrm{E}[\bar{X}] = 4$. $\mathrm{Var}(\bar{X}) = \frac{4+2\cdot1+0+2\cdot1+4}{9} = \frac{12}{9} = 1.333 = 2.667/2$.
(c) For a pair (a, b): $s^2 = (a-b)^2/2$. Equal elements ($s^2 = 0$): 3 pairs; difference 2 ($s^2 = 2$): 4 pairs; difference 4 ($s^2 = 8$): 2 pairs. $\mathrm{E}[s^2] = (0 \cdot 3 + 2 \cdot 4 + 8 \cdot 2)/9 = 24/9 = 2.667 = \sigma^2$.

Problem 9 [B] From a population of $N = 400$ customers with $\sigma = 20$ a sample of $n = 100$ is drawn without replacement. Find the standard error of the mean with and without the finite population correction.

SOLUTION Without the correction: $20/\sqrt{100} = 2$. Correction: $\sqrt{\frac{400-100}{400-1}} = 0.867$, SE $= 2 \cdot 0.867 = 1.73$.

Problem 10 [C] Derive $\mathrm{Var}(\bar{X}) = \sigma^2/n$ and show that $\mathrm{E}[s^2] = \sigma^2$ if $s^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$.

SOLUTION $\mathrm{Var}(\bar{X}) = \frac{1}{n^2}\sum \mathrm{Var}(X_i) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$ (independence).
For s^2: $\sum(X_i - \bar{X})^2 = \sum(X_i - \mu)^2 - n(\bar{X} - \mu)^2$. Taking expectations: $\mathrm{E}\left[\sum(X_i - \bar{X})^2\right] = n\sigma^2 - n \cdot \frac{\sigma^2}{n} = (n-1)\sigma^2$, so $\mathrm{E}[s^2] = \sigma^2$.

Problem 11 [C] The law of large numbers. For $\sigma = 4$ and $\varepsilon = 1$ Chebyshev's inequality gives $P(|\bar{X} - \mu| \geq \varepsilon) \leq \sigma^2/(n\varepsilon^2)$. Estimate the probability for $n = 100$ and compare with the normal approximation. What does the probability tend to as $n \to \infty$?

SOLUTION Chebyshev bound: $16/100 = 0.16$. Normal approximation: SE $= 0.4$, $P(|Z| \geq 2.5) = 0.0124$. The Chebyshev bound is valid for any distribution but very crude. As $n \to \infty$ the right-hand side tends to 0, i.e. $\bar{X}$ converges in probability to μ.

Problem 12 [C] An online poll in the bank's Telegram channel showed 80% satisfied customers with $n = 5000$. Suppose SE ≈ 0.006. Does this mean that the share of satisfied customers among all customers lies within $0.80 \pm 2 \cdot 0.006$? Explain.

SOLUTION No. The standard error describes only the random fluctuation of the sample. The channel's subscribers are not a random sample of customers (self-selection, more loyal customers), so the estimate is *biased*, and increasing n does not remove the bias. The interval 0.80 ± 0.012 may not contain the true share at all.

ANSWERS AND SOLUTIONS · LECTURE 8

The CLT in Practice and Midterm Review

Problem 1 [B] The time (minutes) spent by ten customers on arranging a service: $12,\ 15,\ 15,\ 18,\ 20,\ 22,\ 25,\ 28,\ 30,\ 45$.

(a) Find the mean, median, variance and standard deviation.
(b) Find Q_1, Q_3, IQR and the outlier fences. Are there outliers?
(c) Find the skewness coefficient $Sk = 3(\bar{x} - \text{Me})/s$ and explain the sign. How do the mean and median change without the value 45?

SOLUTION

(a) $\sum x_i = 230$, $\bar{x} = 23$; median $(20 + 22)/2 = 21$. Deviations: $-11, -8, -8, -5, -3, -1, 2, 5, 7, 22$, squares $121, 64, 64, 25, 9, 1, 4, 25, 49, 484$, sum 846. $s^2 = 846/9 = 94$, $s = 9.70$.
(b) Lower half $12, 15, 15, 18, 20$: $Q_1 = 15$; upper $22, 25, 28, 30, 45$: $Q_3 = 28$. $IQR = 13$. Fences: $15 - 19.5 = -4.5$ and $28 + 19.5 = 47.5$. The value 45 is inside the fences — no outliers (although the point is far from the others).
(c) $Sk = 3(23 - 21)/9.70 = 0.62 > 0$: right skew (the mean is to the right of the median). Without 45: $\bar{x} = 20.56$, Me $= 20$; the mean changed more.

Problem 2 [A] $P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.2$. Find $P(A \cup B)$, $P(\overline{A} \cap \overline{B})$, $P(A \mid B)$ and check independence.

SOLUTION $P(A \cup B) = 0.5 + 0.4 - 0.2 = 0.7$; $P(\overline{A} \cap \overline{B}) = 1 - 0.7 = 0.3$; $P(A \mid B) = 0.2/0.4 = 0.5 = P(A)$. Since $P(A)P(B) = 0.2 = P(A \cap B)$, the events are independent.

Problem 3 [B] An online shop: 60% of orders come through the website (5% are returned), 40% through the app (10% are returned).

(a) What is the probability that an order will be returned?
(b) An order was returned. What is the probability that it was placed through the app?

SOLUTION

(a) $0.6 \cdot 0.05 + 0.4 \cdot 0.10 = 0.03 + 0.04 = 0.07$.
(b) $0.04/0.07 = 0.571$.

Problem 4 [A] A group has 10 students (5 boys and 5 girls). A committee of 3 is chosen at random. What is the probability that it contains at least one girl?

SOLUTION Via the complement: $1 - \dfrac{\binom{5}{3}}{\binom{10}{3}} = 1 - \dfrac{10}{120} = 0.9167$.

Problem 5 [B] A customer takes out a subscription with probability 0.2, independently. The offer was

made to 15 customers. Find $\mathrm{E}[X]$, $\mathrm{Var}(X)$, the probability of no subscriptions and the probability of at least four subscriptions.

SOLUTION $X \sim \mathrm{Bin}(15, 0.2)$: $\mathrm{E}[X] = 3$, $\mathrm{Var} = 15 \cdot 0.2 \cdot 0.8 = 2.4$. $P(0) = 0.8^{15} = 0.0352$. $P(X \geq 4) = 1 - P(X \leq 3) = 0.3518$.

Problem 6 [B] There are on average 2.5 requests per minute (Poisson). Find the probability of at most one request per minute and of at least three.

SOLUTION $P(X \leq 1) = e^{-2.5}(1 + 2.5) = 0.0821 \cdot 3.5 = 0.2873$. $P(X \geq 3) = 1 - e^{-2.5}(1 + 2.5 + 3.125) = 1 - 0.0821 \cdot 6.625 = 0.4562$.

Problem 7 [B] A test score is $X \sim N(500, 50^2)$.

(a) Find the share scoring above 580.
(b) Find the share scoring between 450 and 550.
(c) Find the 90th percentile.

SOLUTION

(a) $z = 1.6$, $P = 1 - 0.9452 = 0.0548$.
(b) $z = \pm 1$: 0.6827.
(c) $500 + 1.2816 \cdot 50 = 564.1$.

Problem 8 [B] A game: a win of $+50$ with probability 0.3 and a loss of -10 with probability 0.7. Find the expected result and the standard deviation. Is it worth playing repeatedly?

SOLUTION $\mathrm{E}[X] = 15 - 7 = 8$; $\mathrm{E}[X^2] = 0.3 \cdot 2500 + 0.7 \cdot 100 = 820$; $\mathrm{Var} = 820 - 64 = 756$, $\sigma = 27.5$. The expectation is positive, so over many plays the result is on average $+8$ per game; over a short series the spread is large (3.4 times the mean).

Problem 9 [B] Two returns with correlation $\rho = -0.2$ have $\sigma_1 = 8\%$, $\sigma_2 = 12\%$. Find the risk of a 50/50 portfolio.

SOLUTION $\sigma_p^2 = 0.25 \cdot 64 + 0.25 \cdot 144 + 2 \cdot 0.25 \cdot (-0.2) \cdot 8 \cdot 12 = 16 + 36 - 9.6 = 42.4$; $\sigma_p = 6.51\%$.

Problem 10 [B] The average number of purchases per customer is $\mu = 100$, $\sigma = 24$; a sample of $n = 64$. Find SE, $P(\bar{X} < 95)$ and $P(97 < \bar{X} < 103)$.

SOLUTION $\mathrm{SE} = 24/8 = 3$. $P(\bar{X} < 95) = P(Z < -1.67) = 0.0478$. $P(97 < \bar{X} < 103) = P(|Z| < 1) = 0.6827$.

Problem 11 [B] The share of customers who cancelled the service during a year is $p = 0.25$. A sample of $n = 300$. Find $\mathrm{SE}(\hat{p})$ and $P(\hat{p} > 0.30)$.

SOLUTION $\mathrm{SE} = \sqrt{0.25 \cdot 0.75/300} = 0.025$. $z = 0.05/0.025 = 2$, $P = 0.0228$.

Problem 12 [C] For the conditions of Problem 10 ($\sigma = 24$) find the sample size for which, with probability 95%, the sample mean differs from μ by no more than 4. Then show how n changes if we want to halve the permissible error, and explain this in terms of SE.

SOLUTION $n \geq (1.96 \cdot 24/4)^2 = 11.76^2 = 138.3$, i.e. $n = 139$. For $E = 2$: $n \geq (23.52)^2 = 553.2$, i.e. 554: four times as many. The reason is $\mathrm{SE} = \sigma/\sqrt{n}$: to reduce SE by a factor of 2, $\sqrt{n}$ must be increased by a factor of 2, and n by a factor of 4.

ANSWERS AND SOLUTIONS · LECTURE 9

Estimation and Confidence Intervals

Problem 1 [A] The mean score of a sample of 49 students is 82, $\sigma = 14$. Construct a 95% CI for the mean.

SOLUTION $\text{SE} = 14/7 = 2$; $E = 1.96 \cdot 2 = 3.92$; $\text{CI} = (78.08;\ 85.92)$.

Problem 2 [A] Find from the table the critical value t for: (a) a 95% CI, $n = 10$; (b) a 99% CI, $n = 25$; (c) a 90% CI, $n = 16$.

SOLUTION (a) $\nu = 9$: $t_{0.025,9} = 2.262$. (b) $\nu = 24$: $t_{0.005,24} = 2.797$. (c) $\nu = 15$: $t_{0.05,15} = 1.753$.

Problem 3 [A] A sample of 25 customers: $\bar{x} = 82$ thousand UZS, $s = 10$. Construct a 95% CI for the average bill. Why is t used here rather than z?

SOLUTION $\text{SE} = 10/5 = 2$; $t_{0.025,24} = 2.064$; $E = 4.13$; $\text{CI} = (77.87;\ 86.13)$. We use t because σ is unknown and estimated from a small sample ($n = 25$): the t-distribution accounts for the additional uncertainty of s.

Problem 4 [B] The cost of 16 randomly selected orders: $\bar{x} = 250$ thousand, $s = 40$ thousand. Construct a 99% CI for the average order cost and state what happens to the width when moving to 95%.

SOLUTION $\text{SE} = 40/4 = 10$; $t_{0.005,15} = 2.947$; $E = 29.5$; $\text{CI} = (220.5;\ 279.5)$. At 95% ($t_{0.025,15} = 2.131$): $E = 21.3$, CI $(228.7;\ 271.3)$ — narrower.

Problem 5 [B] Of 500 transactions checked, 85 are flagged as suspicious. Construct 95% and 90% CIs for the share of suspicious transactions.

SOLUTION $\hat{p} = 0.17$; $\text{SE} = \sqrt{0.17 \cdot 0.83/500} = 0.0168$. 95%: $0.17 \pm 1.96 \cdot 0.0168 = (0.137;\ 0.203)$. 90%: $0.17 \pm 1.645 \cdot 0.0168 = (0.142;\ 0.198)$.

Problem 6 [B] Delivery times of 8 orders (minutes): $18, 22, 25, 21, 30, 24, 27, 23$.

(a) Find $\bar{x}$ and s.
(b) Construct a 95% CI for the mean delivery time.
(c) What assumption is needed, and how can it be checked on such data?

SOLUTION

(a) $\sum x = 190$, $\bar{x} = 23.75$; $\sum (x - \bar{x})^2 = 95.5$, $s^2 = 13.64$, $s = 3.69$.
(b) $\text{SE} = 3.69/\sqrt{8} = 1.306$, $t_{0.025,7} = 2.365$, $E = 3.09$; $\text{CI} = (20.66;\ 26.84)$.
(c) For $n = 8$ approximate normality of the population is needed (the CLT does not work). Check: a histogram/box plot without pronounced skewness and outliers (here there are no outliers; the values lie within 2 s of the mean).

Problem 7 [B] To estimate the average response time ($\sigma \approx 20$ s) a margin of error $E = 3$ s is required. Find n for 95% and 99% confidence. Then find n for estimating the share of satisfied customers with $E = 0.03$ at 95% confidence if (a) p is unknown; (b) preliminarily $p \approx 0.6$.

SOLUTION Mean: 95%: $(1.96 \cdot 20/3)^2 = 170.7 \to 171$; 99%: $(2.576 \cdot 20/3)^2 = 294.9 \to 295$. Proportion: (a) $1.96^2 \cdot 0.25/0.03^2 = 1067.1 \to 1068$; (b) $1.96^2 \cdot 0.24/0.03^2 = 1024.4 \to 1025$.

Problem 8 [B] An analyst writes: "With probability 95% the average processing time lies in the interval $(22.8;\ 48.7)$" and "95% of branches have revenue in this interval". Find and correct the errors in both statements.

SOLUTION (1) μ is a constant, so a probability statement refers not to a specific interval but to the method: "the procedure used to build the interval covers μ in 95% of cases"; "we are 95% confident that the interval contains μ" is acceptable. (2) A CI describes the mean, not the distribution of individual observations: the values of individual branches (18…96) go far beyond $(22.8; 48.7)$; for an individual value a prediction interval is needed, which is substantially wider.

Problem 9 [B] How does the width of a CI for a mean (σ known) change if (a) n grows fourfold; (b) the confidence level changes from 95% to 99%; (c) σ decreases by 20%?

SOLUTION Width $\propto z\sigma/\sqrt{n}$. (a) it halves; (b) it grows by a factor of $2.576/1.96 = 1.314$ (by 31%); (c) it decreases by 20%.

Problem 10 [B] In a sample of 80 borrowers, 6 were overdue. Check the conditions, construct the simple 95% CI for the proportion and the Wilson interval. Compare and comment.

SOLUTION $\hat{p} = 0.075$, $n\hat{p} = 6 < 10$ — the condition for the normal approximation is violated. Simple CI: SE $= 0.0294$, $0.075 \pm 0.0577 = (0.017;\ 0.133)$. Wilson interval: $(0.035;\ 0.154)$. The Wilson interval is shifted to the right, as it should be for a proportion near zero (the distribution is skewed), and does not go below 0; for such data it is the one to use.

Problem 11 [C] Derive the CI for μ with known σ from the condition $P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1 - \alpha$, where $Z = (\bar{X} - \mu)/(\sigma/\sqrt{n})$.

SOLUTION $1 - \alpha = P\left(-z_{\alpha/2} < \dfrac{\bar{X} - \mu}{\sigma/\sqrt{n}} < z_{\alpha/2}\right) = P\left(\bar{X} - z_{\alpha/2}\dfrac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z_{\alpha/2}\dfrac{\sigma}{\sqrt{n}}\right)$ — by multiplying by $\sigma/\sqrt{n}$ and moving $\bar{X}$ (the inequality signs flip). The limits are random, μ is fixed.

Problem 12 [C] For a sample of $n = 2$ observations $t_{0.025,1} = 12.706$. Explain why a 95% CI with $n = 2$ is almost useless, while for $n = 100$ ($t = 1.984$) it practically coincides with the z-interval. What role does the estimate s play here?

SOLUTION For $n = 2$ the estimate s rests on one degree of freedom and is extremely unreliable: it may be very small or very large for random reasons. To maintain 95% coverage, the t-distribution introduces a huge multiplier of 12.7 (the interval $\pm 12.7\, s/\sqrt{2}$). For $n = 100$ the value of s is

> determined accurately, so $t \approx z$.

ANSWERS AND SOLUTIONS · LECTURE 10

Hypothesis Testing I: Logic, the p-value and Errors

Problem 1 [A] Formulate H_0 and H_1 (stating the type of test): (a) a manufacturer claims the average pack mass is 500 g, an inspector suspects underweight; (b) a bank wants to check whether the share of overdue payments has changed compared with the previous 6%; (c) a manager believes that a new application form has shortened the average processing time from 15 minutes.

SOLUTION (a) $H_0: \mu = 500$, $H_1: \mu < 500$ (left-sided). (b) $H_0: p = 0.06$, $H_1: p \neq 0.06$ (two-sided). (c) $H_0: \mu = 15$, $H_1: \mu < 15$ (left-sided).

Problem 2 [A] $H_0: \mu = 50$, $H_1: \mu \neq 50$, $\sigma = 10$, sample $n = 100$, $\bar{x} = 52.5$, $\alpha = 0.05$. Carry out the test.

SOLUTION $z = \dfrac{2.5}{10/10} = 2.5$; $p = 2 \cdot 0.0062 = 0.0124 < 0.05$. We reject H_0: the mean differs significantly from 50 (the critical values are ± 1.96, and 2.5 lies beyond them).

Problem 3 [A] A study obtained $p = 0.03$. Which statements are true? (1) The probability that H_0 is true is 3%. (2) If H_0 were true, a result at least as extreme as the observed one would occur in 3% of samples. (3) The effect is large. (4) At $\alpha = 0.05$ H_0 is rejected; at $\alpha = 0.01$ it is not.

SOLUTION (2) and (4) are true. (1) is false: p is the probability of the data given H_0, not of the hypothesis given the data. (3) is false: p does not measure the size of the effect (a small p is possible with a negligible effect and a large n).

Problem 4 [B] $H_0: \mu = 50$, $H_1: \mu \neq 50$. A sample $n = 9$, $\bar{x} = 52$, $s = 6$. Carry out the t-test ($\alpha = 0.05$) and find the p-value.

SOLUTION SE $= 6/3 = 2$, $t = 2/2 = 1.00$, $\nu = 8$. The critical value is $t_{0.025,8} = 2.306$; $|t| < 2.306$. $p = 2P(t_8 > 1) = 0.347$. We do not reject H_0: with such a sample a deviation of 2 units is not proven.

Problem 5 [B] The average delivery time should be 250 minutes. For 16 orders $\bar{x} = 245$, $s = 20$. Test $H_0: \mu = 250$ against $H_1: \mu < 250$.

SOLUTION $t = \dfrac{-5}{20/4} = -1.00$, $\nu = 15$; critical value -1.753; $p = 0.167$. We do not reject H_0: there is no basis to think that delivery is on average faster than the norm.

Problem 6 [B] A company claims that 30% of customers use the app. In a random sample of 500 customers, 170 use the app.

(a) Test $H_0: p = 0.30$ against $H_1: p \neq 0.30$ at $\alpha = 0.05$.
(b) How does the conclusion change for $H_1: p > 0.30$?

(c) Construct a 95% CI and compare with the result of (a).

SOLUTION

(a) $\hat{p} = 0.34$, $\text{SE}_0 = \sqrt{0.3 \cdot 0.7/500} = 0.0205$, $z = 1.95$, $p = 2 \cdot 0.0255 = 0.051 > 0.05$: we do not reject H_0 (a borderline case).

(b) One-sided test: $p = 0.0255 < 0.05$ — we reject. That is why the direction of the hypothesis must be chosen in advance, not after looking at the data.

(c) CI: $0.34 \pm 1.96\sqrt{0.34 \cdot 0.66/500} = (0.298;\ 0.382)$ contains 0.30, which agrees with (a).

Problem 7 [B] Describe Type I and Type II errors in the language of the problem: (a) a bank's monitoring system tests H_0: "the transaction is normal"; (b) testing a new drug, H_0: "the drug has no effect". Which error is more costly in each case and how does this affect the choice of α?

SOLUTION (a) Type I error — blocking a normal transaction (an annoyed customer, checking costs); Type II error — missing fraud (direct losses). Usually the miss is more costly, so a higher α is tolerated (more false alarms for the sake of power). (b) Type I — declaring an ineffective drug effective (harm to patients, costs); Type II — missing a useful drug. Here the Type I error is particularly dangerous, so small α is used (0.01 and below).

Problem 8 [B] $H_0 : \mu = 100$, $\sigma = 15$, $n = 36$, $\alpha = 0.05$ (two-sided).

(a) Find the power of the test at a true $\mu = 105$.

(b) What n is needed for 80% power at $\mu = 105$?

SOLUTION

(a) $\text{SE} = 15/6 = 2.5$, $\delta = 5/2.5 = 2$. Power $= P(Z > 1.96 - 2) + P(Z < -1.96 - 2) = P(Z > -0.04) + 0.00004 = 0.516$.

(b) $n = \left(\dfrac{(1.96 + 0.8416) \cdot 15}{5}\right)^2 = 70.6 \approx 71$.

Problem 9 [B] $n = 64$, $\bar{x} = 103$, $\sigma = 16$. (a) Construct a 95% CI for μ. (b) Without conducting a test, determine the result of a two-sided test at the 0.05 level for $H_0 : \mu = 100$ and for $H_0 : \mu = 98$. (c) Verify this by computing z and p.

SOLUTION (a) $\text{SE} = 2$; $103 \pm 3.92 = (99.08;\ 106.92)$.
(b) 100 is inside the CI — we do not reject; 98 is outside — we reject.
(c) $z = 1.5$, $p = 0.134$ (for 100); $z = 2.5$, $p = 0.0124$ (for 98). The results coincide.

Problem 10 [B] An analyst tested 10 independent hypotheses at the level $\alpha = 0.05$ and found one "significant". (a) What is the probability of at least one false alarm if all 10 null hypotheses are true? (b) What significance level for each test does the Bonferroni correction give?

SOLUTION (a) $1 - 0.95^{10} = 0.401$ — a 40% chance to find "significance" by chance. (b) $\alpha/m = 0.05/10 = 0.005$.

Problem 11 [C] Show that when H_0 is true and the statistic is continuous, the p-value (for a right-sided test) is uniformly distributed on $[0,1]$, i.e. $P(p \le \alpha) = \alpha$ for any α.

SOLUTION Let T be a statistic with a continuous distribution function F under H_0. The right-sided $p = 1 - F(T)$. Then $P(p \le \alpha) = P(F(T) \ge 1-\alpha) = 1 - P(F(T) < 1-\alpha) = 1-(1-\alpha) = \alpha$, since $F(T) \sim U(0,1)$. Hence the rule "reject when $p \le \alpha$" has a Type I error probability of exactly α.

Problem 12 [C] Show that a two-sided z-test at level α rejects $H_0 : \mu = \mu_0$ if and only if μ_0 does not belong to the $(1-\alpha)$ CI $\bar{x} \pm z_{\alpha/2}\sigma/\sqrt{n}$. Then, for $n = 10\,000$, $\bar{x} = 30.3$, $\sigma = 4$, $\mu_0 = 30$, find the p-value and the CI and comment on the practical significance.

SOLUTION The test rejects when $\left|\dfrac{\bar{x}-\mu_0}{\sigma/\sqrt{n}}\right| > z_{\alpha/2}$, i.e. when $|\bar{x}-\mu_0| > z_{\alpha/2}\sigma/\sqrt{n}$ — which is exactly the condition that μ_0 lies outside the interval $\bar{x} \pm z_{\alpha/2}\sigma/\sqrt{n}$.
Numbers: SE $= 0.04$, $z = 7.5$, $p < 10^{-12}$; CI: $30.3 \pm 0.078 = (30.22;\ 30.38)$. The effect is statistically significant, but the CI shows that it is of the order of 0.2–0.4 (about 1% of 30) — practical importance is determined by the business context, not by p.

ANSWERS AND SOLUTIONS · LECTURE 11

Hypothesis Testing II: Comparing Groups, χ^2 and ANOVA

Problem 1 [A] Choose the test (two-sample t, paired t, z for two proportions, χ^2 independence, χ^2 goodness of fit, ANOVA): (a) the average bill in Tashkent, Samarkand and Bukhara; (b) the weight of 20 customers before and after a programme; (c) the share of satisfied customers among men and among women; (d) the relationship between education level (3 categories) and the type of tariff chosen (4 types); (e) the average response time of two groups of operators; (f) whether the distribution of orders over the days of the week is uniform.

SOLUTION (a) ANOVA; (b) paired t; (c) z for two proportions; (d) χ^2 independence; (e) two-sample t (Welch); (f) χ^2 goodness of fit.

Problem 2 [A] Group 1: $n = 40$, $\bar{x} = 50$, $s = 8$; group 2: $n = 50$, $\bar{x} = 46$, $s = 10$. Test $H_0 : \mu_1 = \mu_2$ (H_1 two-sided, $\alpha = 0.05$) and construct a 95% CI for the difference.

SOLUTION $\text{SE} = \sqrt{64/40 + 100/50} = \sqrt{1.6 + 2} = 1.897$; $t = 4/1.897 = 2.11$; $\nu \approx 88$; $p = 0.038 < 0.05$ — we reject H_0. CI: $4 \pm 1.987 \cdot 1.897 = (0.23;\ 7.77)$ — does not contain 0.

Problem 3 [A] The balance of 6 customers (thousand UZS) before and after an advertising campaign: before $40, 52, 38, 45, 60, 48$; after $44, 55, 37, 50, 66, 49$. Test whether the balance increased (two-sided test, $\alpha = 0.05$).

SOLUTION $d = 4, 3, -1, 5, 6, 1$; $\bar{d} = 3$, $s_d = 2.61$, $\text{SE} = 1.065$; $t = 2.82$, $\nu = 5$, $t_{0.025,5} = 2.571$; $p = 0.037$. We reject H_0: the mean increase of 3 thousand is significant. CI: $3 \pm 2.571 \cdot 1.065 = (0.26;\ 5.74)$.

Problem 4 [B] Branch A: $n = 15$, $\bar{x} = 78$, $s = 12$; branch B: $n = 18$, $\bar{x} = 70$, $s = 9$ (customer-service scores). Carry out Welch's test for $H_0 : \mu_A = \mu_B$ and construct a 95% CI for the difference. Compare with the conclusion from the CI.

SOLUTION $\text{SE} = \sqrt{144/15 + 81/18} = \sqrt{9.6 + 4.5} = 3.755$; $t = 8/3.755 = 2.13$; $\nu \approx 25.6$; $t_{0.025} \approx 2.057$; $p = 0.043$. CI: $8 \pm 2.057 \cdot 3.755 = (0.28;\ 15.72)$; 0 is outside the interval, as it should be for $p < 0.05$. The difference is significant, but the interval is very wide: according to the data it could be small (0.3) or large (15.7).

Problem 5 [B] Defect rate on line A: 18 out of 300; on line B: 33 out of 400. Test $H_0 : p_A = p_B$ and construct a 95% CI for the difference.

SOLUTION $\hat{p}_A = 0.060$, $\hat{p}_B = 0.0825$, $\hat{p} = 51/700 = 0.0729$. $\text{SE}_0 = \sqrt{0.0729 \cdot 0.9271 \cdot (1/300 + 1/400)} = 0.0199$. $z = -0.0225/0.0199 = -1.13$, $p = 0.257$ — the difference is not proven. CI: $-0.0225 \pm 1.96 \cdot 0.0199$ (separate error 0.0199)

$= (-0.061;\ 0.016)$.

Problem 6 [B] Of 200 borrowers who received a loan under scheme A, 45 became overdue; of 200 under scheme B — 30. Carry out a χ^2 test of independence (scheme and overdue), $\alpha = 0.05$, and compare with the z-test for two proportions.

SOLUTION Expected counts: $75 \cdot 200/400 = 37.5$ and 162.5 for each scheme. $\chi^2 = \dfrac{7.5^2}{37.5} \cdot 2 + \dfrac{7.5^2}{162.5} \cdot 2 = 1.5 \cdot 2 + 0.346 \cdot 2 = 3.69$, $\nu = 1$, critical value 3.841, $p = 0.0547$ — we do not reject (a borderline case). z-test: $\hat{p}_1 = 0.225$, $\hat{p}_2 = 0.15$, $\hat{p} = 0.1875$, SE $= 0.039$, $z = 1.92$, $p = 0.0547$ — it coincides: for a 2×2 table $\chi^2 = z^2$.

Problem 7 [B] A survey of 350 customers: region (North, Centre, South) and preferred product (loan, deposit, card):

	Loan	Deposit	Card
North	40	30	30
Centre	50	60	40
South	30	20	50

Test the independence of region and product ($\alpha = 0.05$). In which cell is the largest deviation?

SOLUTION Row totals: $100, 150, 100$; column totals: $120, 110, 120$. Expected: North $34.29; 31.43; 34.29$; Centre $51.43; 47.14; 51.43$; South $34.29; 31.43; 34.29$ (all ≥ 5). Contributions $(O-E)^2/E$: $0.95, 0.06, 0.54;\ 0.04, 3.51, 2.54;\ 0.54, 4.16, 7.20$. $\chi^2 = 19.53$, $\nu = 4$, critical 9.488, $p = 0.0006$. We reject independence. The largest contribution: South—Card (7.20; 50 observed against 34.3 expected). $V = 0.167$.

Problem 8 [B] A die was rolled 60 times; faces $1, \ldots, 6$ came up $8, 12, 9, 11, 6, 14$ times. Test the fairness of the die ($\alpha = 0.05$).

SOLUTION $E = 10$ for each face. $\chi^2 = \dfrac{4+4+1+1+16+16}{10} = 4.2$, $\nu = 5$, critical 11.07, $p = 0.52$. There is no basis to consider the die unfair.

Problem 9 [B] One-way ANOVA: 3 groups of $n = 6$ ($N = 18$); means $20, 24, 28$; $MSW = 16$.

(a) Find SSB, MSB and F.
(b) Find the critical value $F_{0.05}$ and draw a conclusion.

SOLUTION

(a) Grand mean 24; $SSB = 6(16 + 0 + 16) = 192$; $MSB = 192/2 = 96$; $F = 96/16 = 6.0$.
(b) $\nu_1 = 2$, $\nu_2 = 15$: $F_{0.05;2,15} = 3.682$; $F > 3.682$ ($p = 0.012$) — we reject H_0: the means are not all equal.

Problem 10 [B] The time to perform an operation in three crews (4 observations each): A: $10, 12, 11, 13$; B: $14, 15, 13, 16$; C: $12, 11, 13, 12$. Build the ANOVA table and draw a conclusion ($\alpha = 0.05$).

SOLUTION Means $11.5;\ 14.5;\ 12.0$; grand mean 12.667. $SSB = 4(1.361 + 3.361 + 0.444) = 20.67$; $SSW = 5 + 5 + 2 = 12$ (by crew: $5, 5, 2$). $MSB = 10.33$, $MSW = 12/9 = 1.333$, $F = 7.75$ with $\nu = (2, 9)$, $F_{0.05;2,9} = 4.256$, $p = 0.011$. We reject H_0. The group variances $1.67, 1.67, 0.67$ — differ by less than a factor of two in s, the condition is met.

Problem 11 [C] For two groups ANOVA is equivalent to a two-sample t-test with a pooled variance: $F = t^2$. Verify on the data of crews A and B from the previous problem.

SOLUTION A: $n = 4$, $\bar{x} = 11.5$, $s^2 = 1.667$; B: $\bar{x} = 14.5$, $s^2 = 1.667$. $s_p^2 = 1.667$, SE $= \sqrt{1.667 \cdot (1/4 + 1/4)} = 0.913$, $t = -3/0.913 = -3.286$, $t^2 = 10.80$. ANOVA: grand mean 13, $SSB = 4 \cdot 1.5^2 \cdot 2 = 18$, $SSW = 5 + 5 = 10$, $F = \dfrac{18/1}{10/6} = 10.8$. They coincide; $p = 0.0167$ in both cases.

Problem 12 [C] In the staff-training data (Lecture 11) $s_{\text{before}} = 6.34$, $s_{\text{after}} = 6.59$, and the correlation between scores before and after is $r = 0.975$. Explain, via the variance of the difference, why the paired test is much more powerful than the independent one. Find the SE for both approaches.

SOLUTION $\operatorname{Var}(d) = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$. Here $s_d^2 = 40.2 + 43.4 - 2 \cdot 0.975 \cdot 6.34 \cdot 6.59 = 2.13$ (computing from the differences gives 2.125). The paired SE $= \sqrt{2.125/8} = 0.515$. An independent analysis ignores the correlation: SE $= \sqrt{40.2/8 + 43.4/8} = 3.23$ — 6.3 times larger. A high correlation between repeated measurements sharply reduces the variance of the difference, so the paired design detects an effect that an independent comparison misses.

ANSWERS AND SOLUTIONS · LECTURE 12

Simple Linear Regression

Problem 1 [A] The regression equation of delivery cost y (thousand UZS) on distance x (km): $\hat{y} = 10 + 2.5x$.

(a) Interpret b_1 and b_0.
(b) Find the prediction at $x = 8$ km and the residual for the observation $(8;\ 32)$.

SOLUTION

(a) $b_1 = 2.5$: each additional kilometre is associated with an increase in the average cost of 2.5 thousand UZS. $b_0 = 10$: the expected cost at zero distance (the fixed part of the tariff, if $x = 0$ is admissible).
(b) $\hat{y} = 10 + 2.5 \cdot 8 = 30$; residual $e = 32 - 30 = 2$ (the actual cost is 2 thousand above the prediction).

Problem 2 [A] Given: $\bar{x} = 40$, $\bar{y} = 100$, $s_x = 10$, $s_y = 30$, $r = 0.6$. Find the regression equation of y on x and R^2.

SOLUTION $b_1 = r\, s_y / s_x = 0.6 \cdot 30/10 = 1.8$; $b_0 = 100 - 1.8 \cdot 40 = 28$; $\hat{y} = 28 + 1.8x$. $R^2 = r^2 = 0.36$.

Problem 3 [A] Study time x (h): $1, 2, 3, 4, 5$; scores y: $50, 55, 65, 70, 85$. Find the regression equation, the prediction at $x = 4$ and the residual for this observation.

SOLUTION $\bar{x} = 3$, $\bar{y} = 65$, $S_{xx} = 10$, $S_{xy} = 85$ (Lecture 6). $b_1 = 8.5$, $b_0 = 65 - 8.5 \cdot 3 = 39.5$, $\hat{y} = 39.5 + 8.5x$. At $x = 4$: $\hat{y} = 73.5$; residual $e = 70 - 73.5 = -3.5$.

Problem 4 [B] For the data of the previous problem find all predictions and residuals, check $\sum e_i = 0$, find SSE, SST, R^2 and s_e.

SOLUTION Predictions: $48,\ 56.5,\ 65,\ 73.5,\ 82$. Residuals: $2,\ -1.5,\ 0,\ -3.5,\ 3$; sum $= 0$. $SSE = 4 + 2.25 + 0 + 12.25 + 9 = 27.5$. $SST = 750$. $R^2 = 1 - 27.5/750 = 0.963$. $s_e = \sqrt{27.5/3} = 3.03$ points.

Problem 5 [B] Apartment area x (m^2): $40, 50, 60, 80, 100$; price y (thousand USD): $38, 45, 50, 68, 84$.

(a) Find the regression equation.
(b) Find R^2 and s_e.
(c) Predict the price of a 70 m^2 apartment. Can the model be used for a 200 m^2 apartment?

SOLUTION

(a) $\bar{x} = 66$, $\bar{y} = 57$, $S_{xx} = 2320$, $S_{xy} = 1800$, $S_{yy} = 1404$. $b_1 = 1800/2320 = 0.776$, $b_0 = 57 - 0.776 \cdot 66 = 5.79$: $\hat{y} = 5.79 + 0.776x$ (each additional m^2 adds 0.78 thousand USD on average).

(b) $SSE = 7.45$ (residuals $1.17;\ 0.41;\ -2.35;\ 0.14;\ 0.62$). $R^2 = 1 - 7.45/1404 = 0.995$; $s_e = \sqrt{7.45/3} = 1.58$ thousand USD.
(c) $\hat{y}(70) = 60.1$ thousand (interpolation). For 200 m^2 the formula gives 161.0, but this is extrapolation far beyond the data (40–100 m^2): the price per metre usually changes nonlinearly with size and class of housing, the prediction cannot be trusted.

Problem 6 [B] For a regression on 22 observations $SST = 500$, $SSE = 120$.

(a) Find SSR and R^2.
(b) Find $|r|$ and s_e.
(c) What can be said about the sign of r?

SOLUTION

(a) $SSR = 500 - 120 = 380$; $R^2 = 380/500 = 0.76$.
(b) $|r| = \sqrt{0.76} = 0.872$; $s_e = \sqrt{120/20} = 2.45$.
(c) The sign of r coincides with the sign of b_1 (it is not determined by R^2).

Problem 7 [B] In the regression of Lecture 12 ($\hat{y} = 8.125 + 3.213x$, x and y in million UZS) express x in thousands of UZS ($x' = 1000x$). How do b_0, b_1, r, R^2 and s_e change?

SOLUTION The predictions remain the same, so $b_0 = 8.125$ and $s_e = 2.195$ do not change. The slope with respect to x': $b_1' = 3.213/1000 = 0.003213$ (million UZS of sales per thousand UZS of advertising). The correlation and R^2 do not depend on linear changes of units: $r = 0.980$, $R^2 = 0.960$.

Problem 8 [B] Using the regression of Lecture 12 an analyst predicts sales for advertising of 25 million UZS and gets $\hat{y} = 88.5$. What is wrong with this prediction? What should he do?

SOLUTION The data range is $x \in [2; 10]$; the value 25 lies far beyond it (extrapolation). A linear relationship found at small spending may not persist (market saturation, diminishing returns). He should either collect data in the needed range or use a nonlinear model and state that the prediction is unreliable.

Problem 9 [B] For the data of Lecture 12 the regression of y on x gives $b_1 = 3.213$. (a) Find the slope of the regression of x on y. (b) Check that the product of the two slopes equals r^2. Can x be found from a predicted y simply by solving the equation $y = b_0 + b_1 x$ for x?

SOLUTION (a) $b_{x|y} = S_{xy}/S_{yy} = 145.67/487.33 = 0.299$. (b) $3.213 \cdot 0.299 = 0.9604 = r^2 = 0.98^2$. Solving the equation for x gives a slope of $1/3.213 = 0.311 \neq 0.299$: this is a different line (OLS minimizes vertical, not horizontal, deviations), so predicting x requires a separate regression.

Problem 10 [C] Derive the OLS estimates: find b_0 and b_1 minimizing $Q = \sum(y_i - b_0 - b_1 x_i)^2$.

SOLUTION $\partial Q/\partial b_0 = -2\sum(y_i - b_0 - b_1 x_i) = 0 \Rightarrow b_0 = \bar{y} - b_1\bar{x}$. $\partial Q/\partial b_1 = -2\sum x_i(y_i - b_0 - b_1 x_i) = 0$. Substituting b_0: $\sum x_i(y_i - \bar{y}) = b_1 \sum x_i(x_i - \bar{x})$, which is equivalent to $S_{xy} = b_1 S_{xx}$, i.e. $b_1 = S_{xy}/S_{xx}$. The second-order conditions hold, since Q is a convex function ($\partial^2 Q/\partial b_1^2 = 2\sum x_i^2 > 0$).

Problem 11 [C] Show that (a) $\sum e_i = 0$; (b) $R^2 = r^2$ in simple regression.

SOLUTION (a) From the condition $\partial Q/\partial b_0 = 0$ we get $\sum e_i = 0$.

(b) $\hat{y}_i - \bar{y} = b_1(x_i - \bar{x})$, so $SSR = b_1^2 S_{xx} = \dfrac{S_{xy}^2}{S_{xx}}$. Then $R^2 = \dfrac{SSR}{SST} = \dfrac{S_{xy}^2}{S_{xx}S_{yy}} = r^2$.

ANSWERS AND SOLUTIONS · LECTURE 13

Regression Inference, Causation and Multiple Regression

Problem 1 [A] In a regression on $n = 27$ observations $b_1 = 2.4$, $\mathrm{SE}(b_1) = 0.6$. Test $H_0 : \beta_1 = 0$ ($\alpha = 0.05$) and construct a 95% CI for β_1.

SOLUTION $t = 2.4/0.6 = 4.0$, $\nu = 25$; $t_{0.025,25} = 2.060$; $p = 0.0005$. We reject H_0. CI: $2.4 \pm 2.060 \cdot 0.6 = (1.16;\ 3.64)$.

Problem 2 [A] The sample correlation is $r = 0.45$ with $n = 30$. Test $H_0 : \rho = 0$ using $t = r\sqrt{n-2}/\sqrt{1-r^2}$ ($\alpha = 0.05$).

SOLUTION $t = \dfrac{0.45\sqrt{28}}{\sqrt{1-0.2025}} = \dfrac{2.381}{0.893} = 2.67$, $\nu = 28$, critical value 2.048, $p = 0.013$. The correlation differs significantly from 0, although the relationship is moderate ($R^2 = 0.20$).

Problem 3 [A] For each statement name a probable lurking variable or an alternative explanation: (a) cities with more churches have more crime; (b) employees who have had more training earn more; (c) firms with high advertising have larger profits.

SOLUTION (a) City size (population). (b) Seniority and position; possibly reverse causation (managers are sent on courses more often). (c) The size and success of the firm; reverse causation (profitable firms spend more on advertising). In all cases the relationship does not prove causation without randomization or control of factors.

Problem 4 [B] Study-time data: $x = 1, 2, 3, 4, 5$, $y = 50, 55, 65, 70, 85$ (Lecture 12): $b_1 = 8.5$, $s_e = 3.03$, $S_{xx} = 10$.

(a) Find $\mathrm{SE}(b_1)$ and test the significance of the slope.
(b) Construct a 95% CI for β_1.
(c) Test $H_0 : \beta_1 = 5$ against $H_1 : \beta_1 \neq 5$.

SOLUTION

(a) $\mathrm{SE}(b_1) = 3.03/\sqrt{10} = 0.957$; $t = 8.5/0.957 = 8.88$, $\nu = 3$, $p = 0.003$ — significant.
(b) $t_{0.025,3} = 3.182$: $8.5 \pm 3.182 \cdot 0.957 = (5.45;\ 11.55)$.
(c) $t = (8.5 - 5)/0.957 = 3.66$, $p = 0.035$ — we reject (5 is outside the CI, consistent).

Problem 5 [B] For the data of the previous problem, at $x_0 = 4.5$ hours find the 95% CI for the mean score and the 95% prediction interval for a new student. Explain the difference.

SOLUTION $\hat{y}_0 = 39.5 + 8.5 \cdot 4.5 = 77.75$. $\sqrt{1/5 + (1.5)^2/10} = 0.6519$, $s_e \cdot 0.6519 = 1.974$. CI for the mean: $77.75 \pm 3.182 \cdot 1.974 = (71.47;\ 84.03)$. Prediction interval: $\sqrt{1 + 0.425} = 1.1937$, $3.03 \cdot 1.1937 = 3.614$; $77.75 \pm 3.182 \cdot 3.614 = (66.25;\ 89.25)$. For a new student the

individual spread ε is added, so the interval is wider.

Problem 6 [B] Apartment data (Lecture 12): $\hat{y} = 5.79 + 0.776x$, $s_e = 1.576$, $n = 5$, $S_{xx} = 2320$, $\bar{x} = 66$.

(a) Test the significance of the slope and construct a 95% CI.
(b) Find the 95% CI for the mean price of 70 m^2 apartments and the prediction interval for the price of a particular 70 m^2 apartment.

SOLUTION

(a) $\mathrm{SE}(b_1) = 1.576/\sqrt{2320} = 0.0327$, $t = 23.7$, $\nu = 3$, $p = 0.0002$. CI: $0.776 \pm 3.182 \cdot 0.0327 = (0.672;\ 0.880)$.
(b) $\hat{y} = 60.10$. For the mean: $\mathrm{SE} = 1.576\sqrt{1/5 + 16/2320} = 0.717$; $60.10 \pm 2.281 = (57.82;\ 62.38)$. Prediction: $\mathrm{SE} = 1.576\sqrt{1.2069} = 1.731$; $60.10 \pm 5.51 = (54.59;\ 65.61)$.

Problem 7 [B] In the output of a regression on $n = 27$ observations it is known that: $SSR = 160$, $SSE = 90$, $b_1 = 2.4$.

(a) Fill in the ANOVA table (ν, MS, F).
(b) Find R^2 and s_e.
(c) Find the t-statistic of the slope and $\mathrm{SE}(b_1)$.

SOLUTION

(a) $SST = 250$; $\nu_{\text{reg}} = 1$, $\nu_{\text{res}} = 25$, $\nu_{\text{total}} = 26$; $MSR = 160$, $MSE = 90/25 = 3.6$; $F = 160/3.6 = 44.4$.
(b) $R^2 = 160/250 = 0.64$; $s_e = \sqrt{3.6} = 1.897$.
(c) $t = \sqrt{F} = 6.67$; $\mathrm{SE}(b_1) = 2.4/6.67 = 0.36$.

Problem 8 [B] Loan applications are submitted at two branches. Approved: branch X — small loans $18/20$, large loans $24/80$; branch Y — small $68/80$, large $5/20$. Find the approval rates by group and overall. Which branch is "better"? Explain the paradox.

SOLUTION Small: X — 90%, Y — 85%; large: X — 30%, Y — 25%. Overall: X — $42/100 = 42\%$, Y — $73/100 = 73\%$. In each group X is better, but overall Y wins — Simpson's paradox: Y mostly receives "easy" small applications (80 out of 100), X difficult large ones (80 out of 100). Branches must be compared within the loan type (or adjusted for the mix of applications).

Problem 9 [B] An online shop introduced free delivery for some customers, choosing those who order more often. A month later their average bill rose by 12% relative to the rest. Can we say that free delivery increased the bill? How should a study be organized to draw a causal conclusion?

SOLUTION No: the groups differ even before the intervention (selection by order frequency is a confounding variable), so the difference may be a result of selection. A randomized A/B test is needed: randomly divide customers into groups with and without free delivery, compare the average bills (a two-sample t-test) and the CI of the difference; or use a before/after comparison with a control group.

Problem 10 [B] A model of weekly sales: $\hat{y} = 20 + 3x_1 - 1.5x_2$ (x_1 — advertising, x_2 — price); $n = 20$, $R^2 = 0.80$.

(a) Interpret the coefficients.
(b) Find the prediction at $x_1 = 8$, $x_2 = 10$.
(c) Find R^2_{adj}.

SOLUTION

(a) At a constant price, +1 in advertising gives on average $+3$ units of sales; at constant advertising, a price increase of 1 reduces sales by 1.5.
(b) $20 + 24 - 15 = 29$.
(c) $R^2_{\text{adj}} = 1 - (1 - 0.8) \cdot \dfrac{19}{17} = 0.776$.

Problem 11 [C] Derive $\text{Var}(b_1) = \sigma^2 / S_{xx}$ using the representation $b_1 = \sum c_i y_i$, where $c_i = (x_i - \bar{x})/S_{xx}$, and the y_i are independent with variance σ^2.

SOLUTION $b_1 = \dfrac{\sum (x_i - \bar{x})(y_i - \bar{y})}{S_{xx}} = \dfrac{\sum (x_i - \bar{x}) y_i}{S_{xx}}$ (since $\sum (x_i - \bar{x}) = 0$). Then $\text{Var}(b_1) = \sum c_i^2 \sigma^2 = \dfrac{\sum (x_i - \bar{x})^2}{S_{xx}^2} \sigma^2 = \dfrac{\sigma^2}{S_{xx}}$. Substituting s_e for σ gives $\text{SE}(b_1) = s_e / \sqrt{S_{xx}}$.

Problem 12 [C] Show that $\text{Var}(\hat{y}_0) = \sigma^2 \left(\dfrac{1}{n} + \dfrac{(x_0 - \bar{x})^2}{S_{xx}} \right)$ and explain where the "1+" in the prediction interval comes from.

SOLUTION $\hat{y}_0 = \bar{y} + b_1 (x_0 - \bar{x})$; $\bar{y}$ and b_1 are uncorrelated (since $\sum (x_i - \bar{x}) = 0$). Therefore $\text{Var}(\hat{y}_0) = \dfrac{\sigma^2}{n} + (x_0 - \bar{x})^2 \dfrac{\sigma^2}{S_{xx}}$. A new observation is $y_0 = \beta_0 + \beta_1 x_0 + \varepsilon_0$ with an independent error ε_0, so the prediction error $y_0 - \hat{y}_0$ has variance $\sigma^2 + \text{Var}(\hat{y}_0)$ — hence the "1" (irreducible individual variability).

ANSWERS AND SOLUTIONS · LECTURE 14

Project Lab: From Question to Conclusion

Problem 1 [A] For the project "What influences the loan amount?", classify the variables and suggest one graph for each: (a) loan amount; (b) loan type (consumer / mortgage / auto); (c) borrower's years of employment; (d) region; (e) loan amount versus years of employment.

SOLUTION (a) continuous quantitative (ratio scale) — histogram/box plot; (b) nominal — bar chart of proportions; (c) quantitative — histogram; (d) nominal — bar chart; (e) two quantitative variables — scatter plot with a regression line.

Problem 2 [A] Choose a method: (a) has the mean application processing time changed after a new system was introduced (for the same 30 employees); (b) are the customer's gender and the chosen card type related; (c) do the mean service ratings differ across 4 branches; (d) by how much does spending grow when income grows by 1 million; (e) do the default rates differ between two lending schemes.

SOLUTION (a) paired t-test; (b) χ^2 test of independence; (c) ANOVA with corrected pairwise comparisons; (d) regression (slope with CI); (e) two-proportion z-test (CI for the difference).

Problem 3 [A] Using the data of the worked mini-project (branch: $n = 10$, $\bar{x} = 4.15$, $s = 0.599$), construct a 95% CI for the mean spending of branch customers.

SOLUTION SE $= 0.599/\sqrt{10} = 0.189$; $t_{0.025,9} = 2.262$; $E = 0.428$; CI $= (3.72;\ 4.58)$.

Problem 4 [B] Two samples of customers: group 1: $n = 40$, $\bar{x} = 52$, $s = 10$; group 2: $n = 45$, $\bar{x} = 47$, $s = 12$.

(a) Find Cohen's d.
(b) Carry out the Welch test and construct a 95% CI for the difference.
(c) Do "statistical" and "practical" significance agree? How would you word the conclusion?

SOLUTION

(a) $s_p = \sqrt{\frac{39 \cdot 100 + 44 \cdot 144}{83}} = 11.1$; $d = 5/11.1 = 0.45$ — a small-to-medium effect.
(b) SE $= \sqrt{100/40 + 144/45} = 2.387$ ($2.50 + 3.20 = 5.70$); $t = 2.09$, $\nu \approx 83$, $p = 0.039$. CI: $5 \pm 1.989 \cdot 2.387 = (0.25;\ 9.75)$.
(c) The difference is significant at the 5% level, but the CI is very wide (from almost zero to almost 10) and the effect is moderate: "a difference exists, but its size is determined imprecisely; whether it matters in practice depends on the business value of 5 units."

Problem 5 [B] Find the flaws in this report: "We surveyed 15 customers chosen by the manager from those who agreed to answer. The mean rating is 4.5 out of 5. The test gave $p = 0.00$, so our service is unquestionably the best in the industry, and the new loyalty program **proved** an increase in satisfaction. Outliers (two ratings of 1) were removed."

SOLUTION (1) The sample is unrepresentative (selected by the manager + self-selection); (2) small size, no CI; (3) it is unknown what was compared and which test was used; $p = 0.00$ is an incorrect notation, it should be $p < 0.001$ with the test named; (4) "unquestionably the best in the industry" does not follow from the data of one company; (5) "proved" is too strong, there is no control group or "before" data; causality is not shown; (6) removing "inconvenient" ratings without justification distorts the result — show the results with and without them; (7) no limitations are described.

Problem 6 [B] An analyst split customers into 20 subgroups (by age, region, channel, etc.) and for each tested whether the mean purchase differs from the bank-wide mean ($\alpha = 0.05$). In 2 subgroups $p < 0.05$, and he published only those.

(a) What is the probability of at least one "significant" subgroup if there are in fact no differences (tests independent)?
(b) What is the p threshold under the Bonferroni correction? How should the results be reported correctly?

SOLUTION

(a) $1 - 0.95^{20} = 0.642$; the expected number of false alarms is $20 \cdot 0.05 = 1$, so two "significant" subgroups are easily explained by chance.
(b) $\alpha^* = 0.05/20 = 0.0025$. One must report all 20 checks, use a correction, and treat the findings as hypotheses to be tested on new data.

Problem 7 [B] The spending data are in columns A (app, A2:A13) and B (branch, B2:B11). Write Excel formulas for: (a) the mean and s of column A; (b) the p-value of the Welch test; (c) the critical t for a 95% CI for the mean of A; (d) $P(Z < 1.5)$. What is the result of (b)?

SOLUTION (a) `=AVERAGE(A2:A13)` $= 5.075$; `=STDEV.S(A2:A13)` $= 0.745$. (b) `=T.TEST(A2:A13,B2:B11,2,3)` $= 0.0042$ (type 3 — unequal variances, 2 — two-sided). (c) `=T.INV.2T(0.05,11)` $= 2.201$. (d) `=NORM.S.DIST(1.5,TRUE)` $= 0.9332$.

Problem 8 [B] For a pilot sample $s = 0.745$. What sample size is needed so that the 95% CI for the mean spending has a margin of error $E = 0.2$ million UZS?

SOLUTION $n \geq (1.96 \cdot 0.745/0.2)^2 = 53.3$, so $n = 54$ (in practice slightly more is taken to allow for the t distribution).

Problem 9 [B] Regression of spending on tenure (app): $\hat{y} = 3.48 + 0.580x$, $\mathrm{SE}(b_1) = 0.0626$, $n = 12$, tenure observed from 1 to 5 years.

(a) Construct a 95% CI for β_1.
(b) Predict the mean spending at a tenure of 3 years. Can a prediction be made for a tenure of 10 years?
(c) Does $b_1 > 0$ mean that tenure increases spending?

SOLUTION

(a) $t_{0.025,10} = 2.228$; $0.58 \pm 2.228 \cdot 0.0626 = (0.44;\ 0.72)$.
(b) $\hat{y} = 3.48 + 1.74 = 5.22$ million UZS. A tenure of 10 years is outside the range 1–5 (extrapo-

lation; the formula would give 9.28, which cannot be considered reliable).

(c) No: the relationship is observational; income, age and habits may be related to tenure (confounding variables).

Problem 10 [B] In the app data, customer 9 had spending 6.3. Suppose 14.5 was mistakenly entered. Recompute the mean, s and the Welch test (branch unchanged). Compare with the original result and state what should be done with such a value.

SOLUTION With the error: $\bar{x} = 5.758$, $s = 2.826$ (instead of 5.075 and 0.745); Welch test: $t = 1.92$, $\nu \approx 12.2$, $p = 0.079$ — the conclusion reverses (at $\alpha = 0.05$ the difference is no longer "significant"). The value 14.5 must be checked against the primary source; if it is confirmed to be an error, correct it and record the correction; if the value is real, show the results with and without it and/or use a robust method (a nonparametric test). The medians are the same in both cases (5.05), since an outlier affects the mean and s but not the median.

Problem 11 [C] An AI assistant reported for the mini-project data: "$t = 2.90$, $\nu = 22$, $p = 0.008$, the difference of means is significant." How can you check this claim? What is wrong?

SOLUTION Recompute independently: from the descriptive statistics SE $= \sqrt{0.745^2/12 + 0.599^2/10} = 0.286$, $t = 0.925/0.286 = 3.23$, $\nu \approx 20.0$ (Welch; for the pooled variance $t = 3.16$, $\nu = 20$, $p = 0.005$), $p = 0.0042$. The AI's claim matches neither variant: 22 degrees of freedom are impossible for $n_1 + n_2 = 22$ (it must be ≤ 20), and t and p are inconsistent with the data. The conclusion of significance happened to survive, but the numbers cannot be used in a report without re-checking.

Problem 12 [C] For an A/B test of conversion the baseline proportion is 30%; you need to detect a rise to 35% ($\alpha = 0.05$, power 80%). (a) Find n per group. (b) What happens if you need to detect a rise to 32%? (c) Why is it important to plan in advance?

SOLUTION (a) $n \approx \dfrac{7.85\,(0.21 + 0.2275)}{0.05^2} = \dfrac{7.85 \cdot 0.4375}{0.0025} = 1374$ in each group. (b) $\dfrac{7.85\,(0.21 + 0.2176)}{0.02^2} \approx 8390$ — an effect 2.5 times smaller requires about 6 times more observations. (c) The minimum meaningful effect, n and the stopping rule are fixed in advance; otherwise "peeking" at the results and stopping at the first $p < 0.05$ inflates the Type I error.

ANSWERS AND SOLUTIONS · LECTURE 15

Final Review and Exam Preparation

Problem 1 [A] Monthly sales (million UZS) of eight managers: $14,\ 18,\ 20,\ 22,\ 25,\ 27,\ 31,\ 63$.

(a) Find the mean, median and standard deviation.
(b) Find Q_1, Q_3, IQR and the outlier fences. Are there outliers?
(c) Which measure of center is preferable and why? How does the mean change without the largest value?

SOLUTION

(a) $\sum x = 220$, $\bar{x} = 27.5$; median $(22+25)/2 = 23.5$; $\sum (x-\bar{x})^2 = 1638$, $s^2 = 234$, $s = 15.30$.
(b) $Q_1 = (18+20)/2 = 19$, $Q_3 = (27+31)/2 = 29$, $IQR = 10$. Fences: $19 - 15 = 4$ and $29 + 15 = 44$. The value 63 exceeds 44 — an outlier.
(c) The median (resistant to the outlier; the mean is pulled upward by 63). Without 63 the mean is 22.43 and the median is 22.

Problem 2 [B] The default rate among borrowers is 4%. A scoring model flags 80% of future defaulters and 10% of reliable borrowers as "risky". A borrower is flagged. What is the probability that he will default? What does the probability become if the borrower is flagged by two independent scoring models with the same characteristics?

SOLUTION $P(\text{default} \mid \text{flag}) = \dfrac{0.8 \cdot 0.04}{0.8 \cdot 0.04 + 0.1 \cdot 0.96} = \dfrac{0.032}{0.128} = 0.25$. Odds: $0.04/0.96 = 0.0417$; $LR = 8$; after two independent flags the odds are $0.0417 \cdot 64 = 2.667$, $P = 2.667/3.667 = 0.727$.

Problem 3 [B] A customer accepts an offer with probability 0.25, independently of others; the offer is made to 12 customers.

(a) Find the expected number of acceptances and the standard deviation.
(b) Find the probability that at least 5 accept.
(c) Find the probability that exactly 3 accept.

SOLUTION $X \sim \text{Bin}(12, 0.25)$.

(a) $\mu = 3$, $\sigma = \sqrt{12 \cdot 0.25 \cdot 0.75} = 1.5$.
(b) $P(X \geq 5) = 1 - P(X \leq 4) = 0.1576$.
(c) $P(X = 3) = C_{12}^{3}\, 0.25^3\, 0.75^9 = 220 \cdot 0.015625 \cdot 0.07508 = 0.2581$.

Problem 4 [A] Test scores $X \sim N(65, 8^2)$. Find $P(X > 75)$, $P(60 < X < 70)$ and the 85th percentile.

SOLUTION $P(X > 75) = P(Z > 1.25) = 0.1056$. $P(60 < X < 70) = P(-0.625 < Z < 0.625) = 0.468$. The 85th percentile: $65 + 1.0364 \cdot 8 = 73.3$.

Problem 5 [B] Number of purchases per customer: $\mu = 2.5$, $\sigma = 1.2$. A sample of $n = 64$ customers. Find $\mathrm{SE}(\bar{X})$ and $P(\bar{X} > 2.7)$. Explain why the normal distribution can be used.

> **SOLUTION** $\mathrm{SE} = 1.2/8 = 0.15$; $z = 0.2/0.15 = 1.33$; $P = 0.0912$. The normal approximation is valid by the CLT ($n = 64 \geq 30$), even if the distribution of the number of purchases per customer is not normal.

Problem 6 [B] A sample of 20 customers: $\bar{x} = 48$ min, $s = 9$. (a) Construct a 95% CI for the mean. (b) Test $H_0 : \mu = 45$ against $H_1 : \mu \neq 45$ ($\alpha = 0.05$). (c) Are the results of (a) and (b) consistent?

> **SOLUTION** (a) $t_{0.025,19} = 2.093$; $\mathrm{SE} = 9/\sqrt{20} = 2.012$; $E = 4.21$; CI $(43.79;\ 52.21)$.
> (b) $t = 3/2.012 = 1.49$, $p = 0.152 > 0.05$ — do not reject.
> (c) Yes: 45 lies inside the CI.

Problem 7 [B] In a sample of 250 customers, 62% use the app. (a) Construct a 95% CI for the proportion. (b) How many customers must be surveyed for the margin of error to be 4 percentage points (with the proportion unknown)?

> **SOLUTION** (a) $\mathrm{SE} = \sqrt{0.62 \cdot 0.38/250} = 0.0307$; $0.62 \pm 0.060 = (0.560;\ 0.680)$. (b) $n \geq 1.96^2 \cdot 0.25/0.04^2 = 600.25$, i.e. 601.

Problem 8 [B] Processing time (min) for 7 employees before and after a new system was introduced: before $30, 28, 35, 40, 25, 33, 38$; after $27, 26, 33, 36, 25, 30, 35$. Test whether processing time decreased (two-sided test, $\alpha = 0.05$) and construct a 95% CI for the mean change.

> **SOLUTION** Differences (after − before): $-3, -2, -2, -4, 0, -3, -3$; $\bar{d} = -2.43$, $s_d = 1.27$, $\mathrm{SE} = 0.481$; $t = -5.05$, $\nu = 6$, $t_{0.025,6} = 2.447$; $p = 0.002$. Reject H_0: the time decreased. CI: $-2.43 \pm 2.447 \cdot 0.481 = (-3.60;\ -1.25)$ min.

Problem 9 [B] A survey of 350 customers: age group and app use.

	Use the app	Do not use
Under 30	60	40
30–45	50	70
Over 45	40	90

Test independence ($\alpha = 0.05$). Which group contributes most to χ^2?

> **SOLUTION** Totals: rows $100, 120, 130$; columns $150, 200$. Expected: $42.86/57.14;\ 51.43/68.57;\ 55.71/74.29$. Contributions: $6.86, 5.14;\ 0.04, 0.03;\ 4.43, 3.32$. $\chi^2 = 19.83$, $\nu = 2$, critical value 5.991, $p < 0.001$. Reject independence. The main contribution comes from the “under 30” group (more users than expected) and the “over 45” group (fewer).

Problem 10 [B] ANOVA for 4 groups of 5 observations each: $SSB = 84$, $SSW = 160$. Complete the table and draw a conclusion ($\alpha = 0.05$).

SOLUTION $\nu_B = 3$, $\nu_W = 16$, $\nu_T = 19$; $MSB = 28$, $MSW = 10$; $F = 2.8$. Critical $F_{0.05;3,16} = 3.239$; $F < 3.239$ ($p = 0.074$). H_0 is not rejected: a difference of means is not established. $\eta^2 = 84/244 = 0.34$ — a noticeable effect, but there are few data (low power).

Problem 11 [B] For $n = 10$ observations: $\bar{x} = 20$, $\bar{y} = 50$, $S_{xx} = 400$, $S_{xy} = 600$, $S_{yy} = 1200$.

(a) Find the regression equation, R^2 and s_e.
(b) Test the significance of the slope and construct a 95% CI for β_1.
(c) Predict y at $x = 25$ and construct a 95% prediction interval.

SOLUTION

(a) $b_1 = 600/400 = 1.5$; $b_0 = 50 - 1.5 \cdot 20 = 20$; $\hat{y} = 20 + 1.5x$. $SSR = b_1 S_{xy} = 900$, $SSE = 1200 - 900 = 300$; $R^2 = 0.75$; $s_e = \sqrt{300/8} = 6.12$.
(b) $\mathrm{SE}(b_1) = 6.12/20 = 0.306$; $t = 4.90$, $\nu = 8$, $p = 0.001$. CI: $1.5 \pm 2.306 \cdot 0.306 = (0.79;\ 2.21)$.
(c) $\hat{y} = 57.5$. $s_e\sqrt{1 + 0.1 + 25/400} = 6.12 \cdot 1.0793 = 6.60$; prediction interval $57.5 \pm 2.306 \cdot 6.60 = (42.3;\ 72.7)$.

Problem 12 [C] A researcher found a positive significant relationship between the number of training sessions held by a bank and the mean income of customers in the region ($r = 0.62$, $p < 0.01$) and proposed increasing the number of training sessions to raise income. Give a critical assessment of the conclusion and suggest how causality could be tested.

SOLUTION Correlation does not imply causation: there may be a lurking variable (the size and development of the region determine both the number of training sessions and income), reverse causality (banks in wealthy regions invest more in education), or chance. A causal conclusion requires an experiment: randomly assign regions/groups of customers to training and control, and compare incomes (two-sample t-test, CI for the difference, effect size); or, with observational data, control for confounders by multiple regression and state the limitations.

Problem 13 [C] A manager wants to know whether customer satisfaction (rating 1–5) differs across three branches and chooses to "run three pairwise t-tests A–B, A–C, B–C at the 0.05 level". Why is this wrong? What should be proposed? What value of α should be used for each pairwise comparison under the Bonferroni correction?

SOLUTION Three tests raise the probability of at least one false alarm to $1 - 0.95^3 = 14.3\%$ (assuming independence). First apply ANOVA (the F-test) to test $H_0 : \mu_A = \mu_B = \mu_C$, and only if it is significant make pairwise comparisons with a correction. Bonferroni: $\alpha^* = 0.05/3 = 0.0167$ for each. It is also worth checking the conditions (similar variances, no strong outliers) and noting that the ratings are ordinal (use a nonparametric test if necessary).

Problem 14 [C] Carry out a mini-analysis. In data on 200 customers the mean spending is $\bar{x} = 4.8$ million, $s = 1.5$. (a) Construct a 95% CI for the mean. (b) The bank claims $\mu = 5$. Test it. (c) What is the effect size $d = (\bar{x} - \mu_0)/s$? (d) Is $n = 200$ sufficient to detect a deviation of 0.2 million with 80% power?

SOLUTION (a) SE $= 1.5/\sqrt{200} = 0.106$; $t_{0.025,199} \approx 1.972$; $E = 0.209$; CI $(4.59;\ 5.01)$. (b) $t = -0.2/0.106 = -1.89$, $\nu = 199$, $p = 0.060$ — at the 5% level we do not reject (5 lies on the edge of the CI; the result is borderline). (c) $d = -0.2/1.5 = -0.13$ — a small effect. (d) The required $n \approx \left(\dfrac{(1.96 + 0.8416) \cdot 1.5}{0.2}\right)^2 = 441.5 \approx 442$, so with $n = 200$ the power is below 80% (about 47%): "no significance" is a consequence of insufficient power, not proof of equality.